

# Tonotopy is not preserved in a descending stage of auditory cortex

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## eLife Assessment

This revised manuscript presents an **important** characterization of mouse auditory cortex receptive field organization, utilizing two-photon imaging of specific subpopulations. They demonstrate a degradation of tonotopic organization from the input to the output neurons. The strength of the evidence is **convincing**.

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## Abstract

Previous studies based on layer specificity suggest that ascending signals from the thalamus to the sensory neocortex preserve spatially organized information, but it remains unknown whether sensory information descending from sensory neocortex to the thalamus also maintains such spatial organization pattern. By focusing on projection specificity, we mapped the tone response properties of two groups of cortical neurons in the primary auditory cortex (A1), based on the relationship between their specific connections to other regions and their function in ascending (thalamocortical recipient, TR neurons) or descending (corticothalamic, CT neurons) auditory information. A clear tonotopic gradient was observed among TR neurons, but not CT neurons. Additionally, CT neurons exhibited markedly higher heterogeneity in their frequency tuning and had broader bandwidth than TR neurons. These results reveal that the information flow descending from A1 to the thalamus via CT neurons is not arranged tonotopically, suggesting that the descending information flow possibly contributes to higher-order feedback processing of diverse auditory inputs.

## Introduction

In the mammalian auditory system, one of the most prominent features is the tonotopic organization — a spatially ordered gradient of neuronal frequency preference (Bizley et al., 2005; Kajikawa et al., 2005; Merzenich et al., 1976; Morel et al., 1993; Nelken et al., 2004; Reale and Imig, 1980; Stiebler et al., 1997; Tani et al., 2018). Clear tonotopic maps have been found in the auditory cortex (AuC) of many mammalian species including humans (Clopton et al., 1974; Humphries et al., 2010), macaque monkeys (Hackett et al., 1998), cats (Lee et al., 2004), ferrets (Bizley et al., 2005), Mongolian gerbils (Budinger et al., 2000; Ohl et al., 2000; Thomas et al., 1993), rats (Polley et al., 2007), and mice (Guo et al., 2012; King et al., 2018). Tonotopy originates in the cochlea and is relayed and preserved throughout all the ascending stages, including the medial geniculate body (MGB) and the AuC (Jasmin et al., 2019; Simon et al., 2009; Smith and Wever, 1949). As one of the most explored pathways in the auditory system, the connections between MGB and AuC play an essential role in the perception of auditory information (Lee, 2013; Ohga et al., 2017; Pardi et al., 2020).

Core fields of the AuC receive their predominant thalamic input from the ventral division of the MGB (MGBv), which confers well-defined frequency tuning arranged into smoothly varying tonotopic gradients (Andersen et al., 1980; Merzenich and Brugge, 1973; Winer et al., 2005). Non-core AuC fields are innervated by non-primary divisions of the MGB and from intracortical inputs originating outside of the AuC (Jones, 2001; Lee and Winer, 2005; Reale and Imig, 1980), which show weak tonotopy and selectivity for processing conspecific communication sounds (Norman-Haignere et al., 2015; Schneider and Woolley, 2013). Feedforward auditory information is conveyed from the MGBv to the AuC, primarily received in layer 4 (L4) but also extending to other layers (Constantinople and Bruno, 2013; Petreanu et al., 2009). Subsequently, it is projected corticofugally to downstream targets from either layer 5 (L5) or layer 6 (L6) of the AuC (Shepherd and Yamawaki, 2021), including feedback information to MGB (Happel et al., 2014; Happel et al., 2010; Homma and Bajo, 2021; Kirchgeßner et al., 2020). Several studies have revealed the sophisticated inputs and the ability to perceive complex sounds of CT neurons in A1 (Clayton et al., 2021; Homma et al., 2017), suggesting the descending information flow in the core fields of the AuC possibly contributes to higher-order feedback processing of diverse auditory inputs.

It is also known that the tonotopic gradient is well preserved across all layers within the A1 (Guo et al., 2012; Montes-Lourido et al., 2021; Tischbirek et al., 2019; Winkowski and Kanold, 2013). However, all of these previous discoveries in tonotopic mapping are based on layer specificity. While the laminar structure is closely associated with functional projections in the AuC, these are not precisely identical. Neurons within the same layer can exhibit diversity in their molecular, morphological, physiological, and connectional features (Triarhou, 2021; Yarmohammadi et al., 2014), and therefore, categorizing cortical neurons based on their projection specificity can provide a deeper fundamental understanding of their functional organization. The A1-MGB projecting neurons play an essential role in the feedforward-feedback loop of the thalamus-neocortex interplay for integrated sound information processing flow (Happel et al., 2010) and the perception of complex sounds (Clayton et al., 2021; Homma et al., 2017). However, due to technical difficulties, there has been limited understanding of the functional properties of these specific corticothalamic projecting neurons (Antunes and Malmierca, 2021; Winer et al., 2001), because they are located in deeper layers of the cortex and are more difficult to access by means of commonly-used, single-cell-resolving recording techniques such as conventional two-photon (2P) microscopy (Kobat et al., 2009; Oheim et al., 2001; Takasaki et al., 2020; Tischbirek et al., 2015).

Recent work has established a rabies-based retrograde labeling method with pathway specificity (Sun et al., 2019 [↗](#); Zhu et al., 2020 [↗](#)) to circumvent the neuropil contamination problem associated with conventional 2P imaging for CT neurons. Here, using viral tracing and 2P  $\text{Ca}^{2+}$  imaging of awake head-fixed mice, we systematically revisited the functional organization in the A1, specifically targeting two types of neurons with projection specificity: TR (Zingg et al., 2017 [↗](#)) and CT neurons (Gu et al., 2023 [↗](#)), akin to the input and output of the AuC. The projection specificity-based approach provides detailed insight into the differences in population activity and frequency response-related topographic organization between neurons with different connectivity specificities.

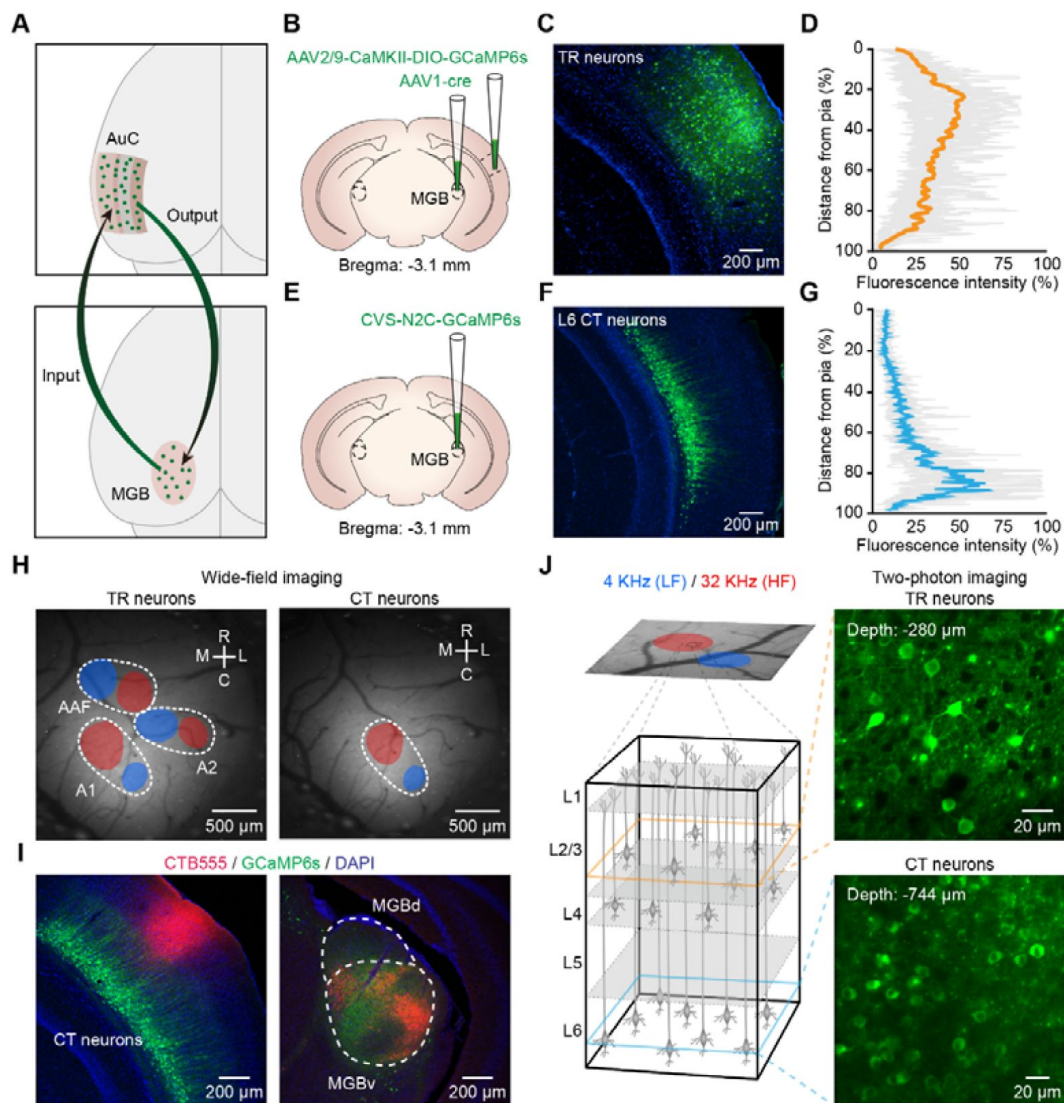
## Results

### Pathway-specific labeling of TR and CT neurons

To label TR neurons, we used an anterograde trans-synaptic tracing strategy by injecting a Cre-recombinase-expressing AAV into the MGB, then injecting a second AAV carrying Cre-dependent GCaMP6s (AAV2/9-CaMKII-DIO-GCaMP6s) in the AuC (**Figure 1A** [↗](#) and **B** [↗](#)) (Zingg et al., 2017 [↗](#)). Neurons labeled with AAV-GCaMP6s were observed across all layers of the AuC (**Figure 1C** [↗](#) and **D** [↗](#)). To label CT neurons, we performed retrograde tracing with a rabies virus CVS-N2c-derived vector (CVS-GCaMP6s) by injecting it into the MGB (**Figure 1E** [↗](#)). Neurons labeled with CVS-GCaMP6s were restricted to AuC L6 (**Figure 1F** [↗](#) and **G** [↗](#)) and were then imaged by 2P  $\text{Ca}^{2+}$  imaging, in line with our recent work (Gu et al., 2023 [↗](#)). Whole-cell patch-clamp recordings and in vivo 2P imaging showed no change in electrophysiological, morphological, or functional characteristics of CVS-labeled CT neurons (Gu et al., 2023 [↗](#)). To locate A1, we conducted wide-field imaging in awake GCaMP6s-labeled mice, and defined that the low-frequency tone (4 kHz) elicited spatially restricted responses in the regions referred to as “low-frequency (LF) area”, and that the high-frequency tone (32 kHz) elicited responses in the regions referred to as “high-frequency (HF) area” (**Figure 1H** [↗](#)). In AAV-GCaMP6s labeled mice, we identified the locations of A1, anterior auditory field (AAF), and secondary auditory field (A2) based on the known tonotopy in mice (**Figure 1H** [↗](#), left panel) (Issa et al., 2014 [↗](#); Liu et al., 2019 [↗](#)).

It is worth noting that, in CVS-GCaMP6s-labeled (i.e., CT neurons) mice, pure tone stimulation elicited wide-field signals in only one auditory region that exhibited a tonotopic gradient (**Figure 1H** [↗](#), right panel). To identify this region in AuC, we injected a retrograde tracing indicator cholera toxin subunit B conjugated with Alexa 555 (CTB-555) into this region, which resulted in clearly visible CTB-555-labeled cell bodies in the MGBv (**Figure 1I** [↗](#)), suggesting that CVS-GCaMP6s-labeled neurons were located in A1 (Rothschild et al., 2010 [↗](#)). When performing 2P imaging, the average power delivered to the brain was in the range of 30–120 mW, depending on the depth of imaging. To determine whether neurons labeled with the CVS retained normal response properties after deep 2P imaging—with higher laser power than that used for superficial layer imaging—we observed the broadband noise-evoked (BBN-evoked) responses of these neurons over time (days) (**Figure 1—figure supplementary 1A** [↗](#)). At 7 and 13 days after injection, we found that CVS-labeled neurons had completely normal response properties (**Figure 1—figure supplementary 1B** [↗](#) and **C** [↗](#)).

Next, we performed  $\text{Ca}^{2+}$  imaging of TR or CT neurons from A1 identified by wide-field imaging (**Figure 1J** [↗](#)). High-titer AAV1-Cre virus has been reported to anterogradely label postsynaptic neurons and also retrogradely label presynaptic neurons (Zingg et al., 2017 [↗](#)). To minimize this labeling ambiguity, we restricted the 2P imaging depth of TR neurons to 200–400  $\mu\text{m}$ , without imaging the neurons in L5 and L6, which were reciprocally connected with the MGB (Harris and Mrsic-Flogel, 2013 [↗](#)). These results suggest the feasibility of our approach to label and record population  $\text{Ca}^{2+}$  signals with pathway specificity at single-cell resolution in awake mice.



**Figure 1.**

### Imaging TR or CT neurons in A1 of awake mice.

(A) Cartoon illustration of the auditory thalamocortical and corticothalamic circuits. (B) Schematic diagram of the injection site of AAV2/1-Cre in the MGB and AAV2/9-CaMKII-DIO-GCaMP6s in AuC. (C) Coronal slice showing AAV-GCaMP6s expression in the AuC on day 21. (D) Fluorescence intensity of TR neurons in AuC with distance from pia (0%) to the L6/WM border (100%). (E) Schematic diagram of the injection site of CVS-GCaMP6s in the MGB. (F) Coronal slice showing the CVS-GCaMP6s retrogradely labeled neurons in the AuC on day 7. (G) Fluorescence intensity of CT neurons in AuC with distance from pia (0%) to the L6/WM border (100%). (H) Left: wide-field imaging in AAV-GCaMP6s-expressing mouse (TR neurons), fluorescence response to pure-tone stimulation with 4 kHz (blue area) and 32 kHz (red area). White dotted lines outline the A1, A2, and AAF boundaries. Right: wide-field imaging in CVS-GCaMP6s-expressing mouse (CT neurons). (I) Left: a fluorescent micrograph of a coronal slice after CTB-555 loading guided by 2P imaging into A1 in a CVS-GCaMP6s-expressing mouse. Right: micrograph of a coronal slice of the MGB from the same mouse. CTB-555 retrogradely labeled neurons were mainly concentrated in the MGBv. (J) Left: wide-field imaging (in a GCaMP6s-expressing mouse) in A1, fluorescence response to pure-tone stimulation with 4 kHz (low-frequency (LF) area) and 32 kHz (high-frequency (HF) area), same abbreviation for all subsequent figures. Inset panels outlined by dashed boxes show the magnified views of 2P imaging of TR or CT neurons. Right: examples of 2P images of TR and CT neurons *in vivo*, respectively.

## CT neurons exhibit no tonotopic gradient

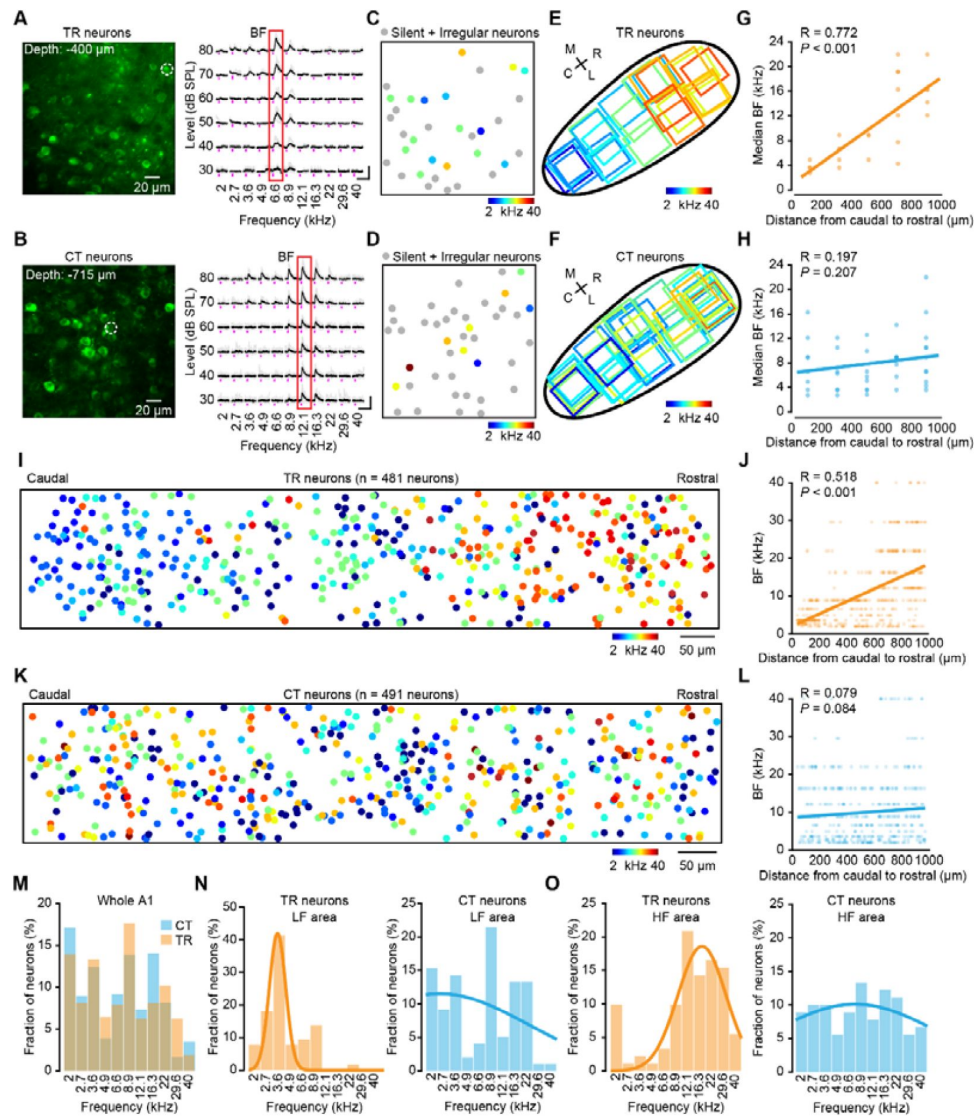
To study tonal response profiles, we delivered 330 pure tones at each focal plane (11 frequencies ranging from 2–40 kHz, 6 attenuation levels, 5 repetitions). We determined the best frequency (BF) (Guo et al., 2012) of all tuned TR (Figure 2A) or CT neurons (Figure 2B) at each 2P imaging focal plane. In total, we imaged 1041 TR neurons in 5 animals. 46% were “tuned” neurons, 41% were “irregular” neurons, and 13% were “silent” neurons (Figure 2—figure supplementary 2A, the definitions of “irregular”, “tuned”, and “silent”, see METHOD DETAILS). For CT neurons, we imaged a total of 2721 neurons from 10 animals. 18% were “tuned” neurons, 57% were “irregular” neurons and 25% were “silent” neurons (Figure 2—figure supplementary 2B).

Based on the BFs, we determined the median BF of all tuned neurons in each 2P focal plane (e.g., Figure 2C and D) and related them to the position within the wide-field imaging areas (Figure 2E and F), which were then used for computing the tonotopic gradient. For TR neurons, we observed a clear tonotopic axis in the organization of median BF ( $R = 0.77$ ,  $P < 0.001$ ; 2.13 octaves/mm; Figure 2G). However, no such tonotopic gradient was observed in CT neurons ( $R = 0.20$ ,  $P = 0.21$ ; 0.21 octaves/mm; Figure 2H). Plotting the relative positions of all individual tuned TR or CT neurons in each mouse revealed a significant low-to-high frequency gradient from caudal to rostral for TR neurons ( $R = 0.52$ ,  $P < 0.001$ ; 1.76 octaves/mm; Figure 2I and J), but an absence of tonotopy in CT neurons ( $R = 0.079$ ,  $P = 0.084$ ; 0.37 octaves/mm; Figure 2K and L). These results, at single-neuron resolution, confirmed that the TR neuronal population does possess tonotopy, as previous reports have shown on a coarse scale (Kalatsky et al., 2005; Wu et al., 2006).

The fraction of different BF responsive neurons among TR and CT neurons showed that both TR and CT neurons displayed no obvious pattern throughout A1 (Figure 2M). When categorized into LF or HF areas, higher proportions of TR neurons responded to low-frequency sounds in the LF areas or high-frequency sounds in the HF areas, whereas CT neurons showed no frequency preference in either area (Figure 2N and O). Note the obvious discrepancy in that the 2P imaging data here shows no tonotopic gradient of L6 CT neurons, but widefield epifluorescence imaging data shows the presence of a tonotopic gradient from the same mice (Figure 1H), which could be largely due to differences in the imaging methods and will be discussed later on.

To validate the reliability of our imaging/analysis approach, we performed control experiments that imaged the general excitatory neurons in superficial layers (Figure 2-figure supplementary 3), and the results showed a clear tonotopic gradient, which was consistent with previous findings (Bandyopadhyay et al., 2010; Romero et al., 2020; Rothschild et al., 2010; Tischbirek et al., 2019). Moreover, to definitively disentangle projection-specific properties from general layer-specific properties, we first analyzed TR neurons to see if response properties varied by depth within the superficial layers. We found no significant differences in the fraction of tuned neurons, field IQR, or BWmax between TR neurons in L2/3 versus those in L4 (Figure 2-figure supplementary 4). This suggests a degree of functional homogeneity within the thalamorecipient population across these layers. To directly test if projection identity confers distinct functional properties within the same cortical layers, we performed the crucial control of comparing TR neurons to their neighboring non-TR neurons. Our results show that TR neurons are significantly more likely to be tuned to pure tones than their neighboring non-TR excitatory neurons (Figure 2-figure supplementary 5). This finding provides direct evidence that a neuron’s long-range connectivity, and not just its laminar location, is a key determinant of its response properties.





**Figure 2.**

### Tonotopic gradients of TR versus CT neurons.

(A) Two-photon image of the focal plane used to image TR neurons in A1. The dashed circle indicates the tuned neurons. Fluorescence traces of neurons in the right panel are indicated by the dashed circle ordered according to sound frequency and level. The five traces associated with the five repeats of each stimulus are plotted in gray. The average calcium signals are plotted in black. The red outline marks the BF response of the neuron. (B) Same as panel (A) but for CT neurons in A1. (C) BF map of TR neurons. (D) BF map of CT neurons. (E) Schematic illustrating the recording locations of individual fields of view of TR neurons in A1. Outlines are color-coded according to the median BF response in the respective fields of view, with their BFs (kHz) color-coded by the scheme below.  $N = 23$  focal planes, from 5 mice. (F) Same as panel (E) but for CT neurons.  $N = 40$  focal planes, from 10 mice. (G)  $BF_{\text{median}}$  plotted against distance along the tonotopic axis for TR neurons. Scatter plots showing the correlation between the cellular  $BF_{\text{median}}$  values measured in the fields of view with 2P imaging and the corresponding extrapolated brain surface frequencies determined by wide-field imaging of TR neurons. (H) Same as panel (G) but for CT neurons. (I) Reconstruction of the relative spatial locations of tuned TR neurons that covered the whole A1 area and color-coded according to each neuron's BF.  $N = 481$  neurons from 5 mice. (J) Plots of TR neurons' BF and their relative distances along the caudal-to-rostral axis. (K) Same as panel (I) but for CT neurons.  $N = 491$  neurons from 10 mice. (L) Same as panel (J) but for CT neurons. (M) Distribution histogram of BF from all neurons in the dataset (TR:  $N = 481$  neurons; CT:  $N = 491$  neurons). (N) Distribution histogram of BF neurons that were identified in the LF area by wide-field imaging (TR:  $N = 116$  neurons; CT:  $N = 98$  neurons). (O) Distribution histogram of BF neurons that were identified in HF area by wide-field imaging (TR:  $N = 91$  neurons; CT:  $N = 90$  neurons).

## CT neurons exhibit high heterogeneity in tuning preference

Next, we performed an analysis of the BF spatial distribution within each focal plane. Examination of BF distribution among TR or CT neurons revealed that TR neurons exhibited similar frequency preference in HF or LF areas (**Figure 3A**), whereas CT neurons displayed high heterogeneity in their frequency preferences, regardless of LF or HF areas (**Figure 3B**).

To quantify the degree of BF heterogeneity, we examined BF spatial distribution at the focal plane, local, and nearest-neighboring scales. To this end, the inter-quartile range of BF ( $IQR_{BF}$ , in octaves) was calculated for all tuned neurons in each focal plane (**Figure 3C** and **D**). Quantitative analysis showed that CT neurons had higher  $IQR_{BF}$  than TR neurons across all focal planes (**Figure 3C**). Similarly, evaluation of LF or HF areas indicated that CT neurons had significantly higher  $IQR_{BF}$  than TR neurons in LF areas, but showed no significant difference from TR neurons in HF areas at the focal plane level (**Figure 3D**).

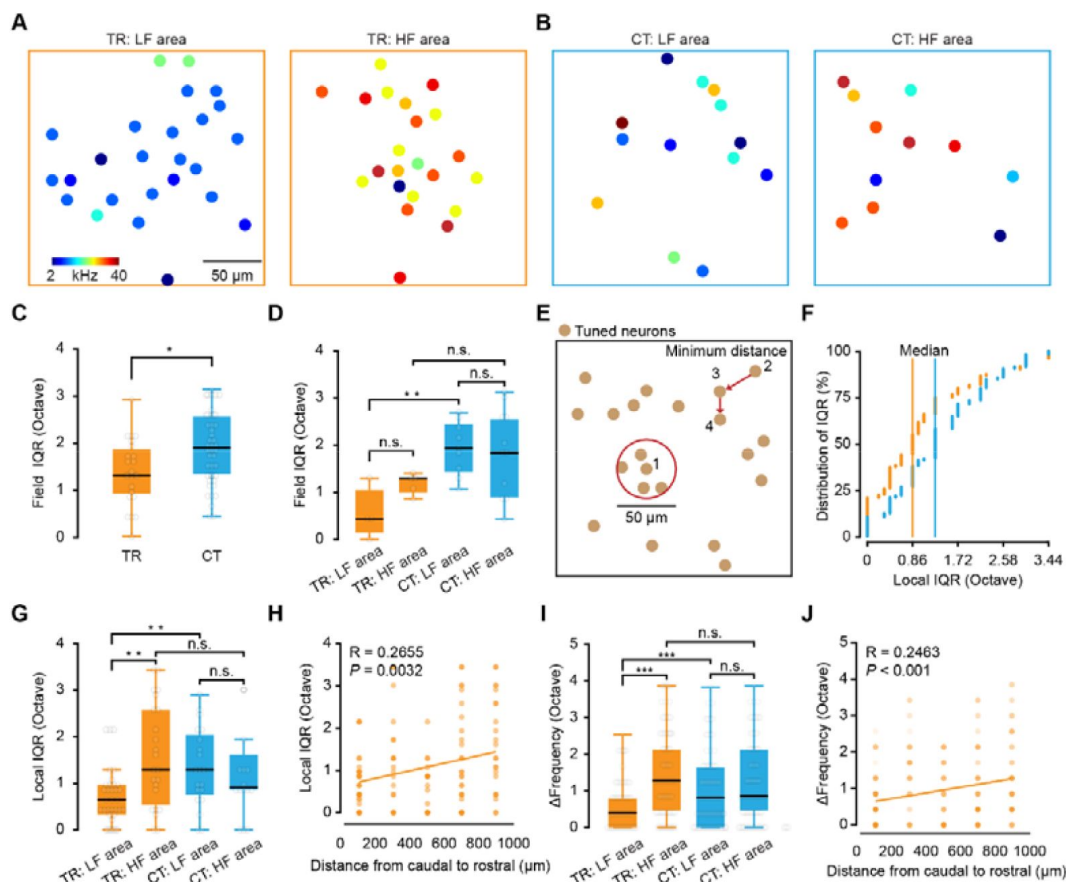
To assess tone response heterogeneity at the local scale, we then calculated  $IQR_{BF}$  for all tuned neurons within a 25  $\mu m$  radius (Winkowski and Kanold, 2013; Zeng et al., 2019) around each randomly selected CT or TR neuron (**Figure 3E–G**). The results revealed that TR neurons also exhibited lower median  $IQR_{BF}$  than CT neurons at the local scale (**Figure 3F**). Consistent with the focal plane scale, CT neurons had significantly higher local  $IQR_{BF}$  than TR neurons in LF areas (**Figure 3G**). Unexpectedly, the local  $IQR_{BF}$  of TR neurons in HF areas was significantly higher than that of TR neurons in LF areas (**Figure 3G**). Further analysis across local regions revealed that the local  $IQR_{BF}$  of TR neurons followed a low-to-high gradient along the caudal-to-rostral axis ( $R = 0.2655$ ,  $P = 0.0032$ ; **Figure 3H**).

The analysis of  $\Delta$ frequency (BF variability) (Bandyopadhyay et al., 2010; Zeng et al., 2019) between nearest-neighboring TR or CT neurons was consistent with results of local  $IQR_{BF}$  analysis (**Figure 3I**). That is,  $\Delta$ frequency followed a clear gradient in the organization of TR neurons along the caudal-to-rostral axis ( $R = 0.2463$ ,  $P < 0.001$ ; **Figure 3J**). Moreover, TR neurons, but not CT neurons, displayed high  $IQR_{BF}$  in HF areas compared to LF areas (**Figure 3G** and **I**). These results suggest the relative heterogeneity of frequency preference among CT neurons but homogeneity among TR neurons.

## Receptive field properties of TR versus CT neurons

Given the above differences in the functional organization of frequency preference between TR and CT neurons, we next focused our analysis on the frequency responsive areas (FRAs) of individual neurons, as the FRAs directly reflect neuronal auditory selectivity essential for sound processing (Sadagopan and Wang, 2008). For this analysis, all tuned neurons were categorized into either a V-shaped (decreasing frequency selectivity with increasing intensity), I-shaped (narrow, level-tolerant tuning), or O-shaped (non-monotonic) FRA (**Figure 4A** and **B**). We found that a higher proportion of CT neurons was associated with V-shaped FRAs than TR neurons, whereas the proportion of CT neurons associated with I-shaped FRAs was lower than that of TR neurons (**Figure 4C**). By contrast, no difference was detected in the proportions of CT and TR neurons associated with O-shaped FRAs (**Figure 4C**).

A key feature of neuronal tuning curves is their sharpness. Common measures of sharpness in V shape are half-peak bandwidth and the “quality factor” (Q), which was obtained by dividing the BF of the neuron by a measure of tuning (Micheyl et al., 2013), the width of the tuning curve at half-peak in this study. This analysis showed that the half-peak bandwidth of CT neurons was significantly wider than that of TR neurons (**Figure 4D**), while the Q value of CT neurons was lower than that of TR neurons, especially in the HF areas (**Figure 4E** and **F**). Since higher Q

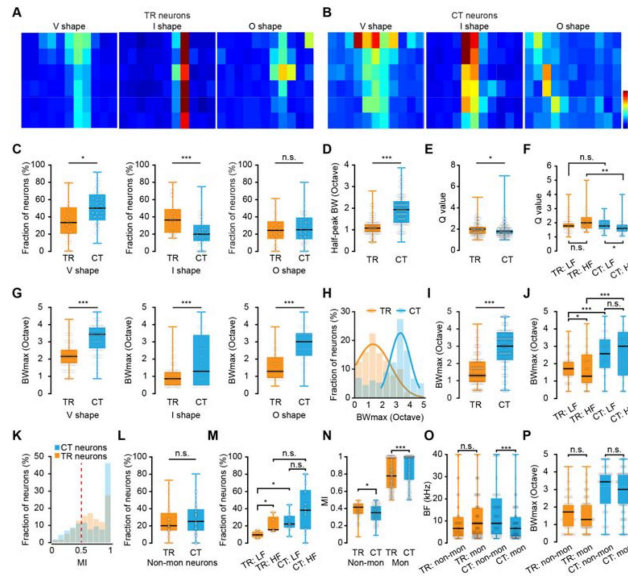


**Figure 3.**

### Heterogeneity of frequency preference in TR versus CT neurons.

(A) Spatial distribution of all tone-selective neurons in an example imaging plane of TR neurons from LF (left) and HF (right) imaging areas. (B) Same as panel (A) but for CT neurons. BFs (kHz) color code is on panel (A). (C) Comparison of the field (200  $\mu$ m) IQR of TR and CT.  $N = 23$  focal planes from 5 mice for TR;  $N = 40$  focal planes from 10 mice for CT. "TR": 1.29\0.91–1.88 octaves (median\25–75% percentiles, same notation for all subsequent data), "CT": 1.94\1.29–2.58 octaves.  $P = 0.016$ , two-sided Wilcoxon rank-sum test,  $*P < 0.05$ ,  $**P < 0.01$ , and  $***P < 0.001$ , same statistics for boxplots. (D) Comparison of the field IQR of TR and CT neurons in LF and HF areas. LF area:  $N = 4$  focal planes from 3 mice for TR;  $N = 9$  focal planes from 5 mice for CT. HF area:  $N = 5$  focal planes from 5 mice for TR;  $N = 11$  focal planes from 10 mice for CT. "TR: LF area": 0.43\0.22–0.86 octaves, "TR: HF area": 1.29\1.02–1.32 octaves, "CT: LF area": 1.94\1.45–2.42 octaves, "CT: HF area": 1.83\0.94–2.58 octaves.  $P$  (TR: LF area, HF area) = 0.11,  $P$  (CT: LF area, HF area) = 0.82,  $P$  (LF area: TR, CT) = 0.0084,  $P$  (HF area: TR, CT) = 0.25. (E) Cartoon showing how the local and minimum percentages were computed. (F) Cumulative percentage plot displaying local (25  $\mu$ m radius) IQR around each neuron of TR and CT. Median local IQR<sub>BF</sub> of TR neurons = 0.86 octaves,  $N = 120$  local planes from 5 mice; median local IQR<sub>BF</sub> of CT neurons = 1.29 octaves,  $N = 105$  local planes from 10 mice. (G) Comparison of the local (25  $\mu$ m radius) IQR of TR and CT neurons in LF and HF areas. LF area:  $N = 34$  local planes for TR;  $N = 20$  local planes for CT. HF area:  $N = 23$  local planes for TR;  $N = 16$  local planes for CT. "TR: LF area": 0.65\0.32–0.97 octaves, "TR: HF area": 1.29\0.54–2.58 octaves, "CT: LF area": 1.29\0.75–2.04 octaves, "CT: HF area": 0.91\0.86–1.61 octaves.  $P$  (TR: LF area, HF area) = 0.0021,  $P$  (CT: LF area, HF area) = 0.69,  $P$  (LF area: TR, CT) = 0.0057,  $P$  (HF area: TR, CT) = 0.34. (H) Local IQR<sub>BF</sub> plotted against distance along the tonotopic axis for TR neurons.  $R = 0.2655$ ,  $P = 0.0032$ . (I) Comparison of the minimum distance  $\Delta$  Frequency of TR and CT neurons in LF and HF area. LF area:  $N = 116$  paired neurons for TR;  $N = 87$  paired neurons for CT. HF area:  $N = 91$  paired neurons for TR;  $N = 78$  paired neurons for CT. "TR: LF area": 0.43\0.00–0.86 octaves, "TR: HF area": 0.86\0.00–1.72 octaves, "CT: LF area": 0.43\0.00–1.29 octaves, "CT: HF area": 0.43\0.00–1.29 octaves.  $P$  (TR: LF area, HF area) = 4.25e-04,  $P$  (CT: LF area, HF area) = 0.79,  $P$  (LF area: TR, CT) = 0.18,  $P$  (HF area: TR, CT) = 0.025. (J)  $\Delta$  Frequency plotted against distance along the tonotopic axis for TR neurons.  $R = 0.2463$ ,  $P < 0.001$ .





**Figure 4.**

### Receptive field properties of TR versus CT neurons.

(A) One representative FRA from a specific type of TR neuron. Types are V shape, I shape, and O shape. See text for further details. The X-axis represents 11 pure tone frequencies, and the Y-axis represents 6 sound intensities. (B) One representative FRA from a specific type of CT neuron. (C) Comparison of fraction of neurons from V, I, and O shaped neurons in TR and CT. (D) Comparison of half-peak bandwidth of V-shaped TR and CT neurons.  $N = 190$  neurons from 5 mice for TR;  $N = 249$  neurons from 10 mice for CT. "TR":  $1.08 \backslash 0.86-1.29$  octaves, "CT":  $1.94 \backslash 1.29-2.37$  octaves.  $P = 4.86e-29$ . (E) Comparison of Q value (bandwidth of BF/half bandwidth) of V-shaped TR and CT neurons.  $N = 190$  neurons from 5 mice for TR;  $N = 249$  neurons from 10 mice for CT. "TR":  $2.00 \backslash 1.50-2.29$ , "CT":  $1.78 \backslash 1.50-2.00$ .  $P = 0.035$ . (F) Comparison of the Q value of V-shaped TR and CT neurons in LF and HF areas. LF area:  $N = 56$  neurons for TR;  $N = 49$  neurons for CT. HF area:  $N = 37$  neurons for TR;  $N = 41$  neurons for CT. "TR: LF area":  $1.78 \backslash 1.60-2.00$ , "TR: HF area":  $2.00 \backslash 1.58-2.43$ , "CT: LF area":  $1.78 \backslash 1.50-2.22$ , "CT: HF area":  $1.60 \backslash 1.33-1.88$ .  $P$  (TR: LF area, HF area) = 0.20,  $P$  (CT: LF area, HF area) = 0.032,  $P$  (LF area: TR, CT) = 0.88,  $P$  (HF area: TR, CT) = 0.0032. (G) Comparison of maximum bandwidth ( $BW_{max}$ ) from V, I, and O shaped neurons in TR and CT. V shape:  $N = 190$  neurons for TR,  $N = 249$  neurons for CT; "TR":  $2.15 \backslash 1.72-2.58$  octaves, "CT":  $3.44 \backslash 2.58-3.87$  octaves;  $P = 4.18e-29$ . I shape:  $N = 174$  neurons for TR,  $N = 112$  neurons for CT; "TR":  $0.86 \backslash 0.43-1.29$  octaves, "CT":  $1.29 \backslash 0.43-3.44$  octaves;  $P = 4.66e-05$ . O shape:  $N = 117$  neurons for TR,  $N = 130$  neurons for CT; "TR":  $1.29 \backslash 0.86-2.15$  octaves, "CT":  $3.01 \backslash 2.15-3.44$  octaves;  $P = 4.30e-19$ . (H) Distribution histogram of  $BW_{max}$  from all neurons in the dataset. (I) Comparison of the  $BW_{max}$  of TR and CT neurons.  $N = 481$  neurons from 5 mice for TR,  $N = 491$  neurons from 10 mice for CT; "TR":  $1.29 \backslash 0.86-2.15$  octaves, "CT":  $3.01 \backslash 2.15-3.87$  octaves;  $P = 1.89e-51$ . (J) Comparison of the  $BW_{max}$  of TR and CT neurons in LF and HF areas. LF area:  $N = 116$  neurons for TR;  $N = 98$  neurons for CT. HF area:  $N = 91$  neurons for TR;  $N = 90$  neurons for CT. "TR: LF area":  $1.72 \backslash 1.29-2.15$ , "TR: HF area":  $1.29 \backslash 0.86-2.47$ , "CT: LF area":  $2.58 \backslash 1.72-3.44$ , "CT: HF area":  $3.01 \backslash 1.29-3.87$ .  $P$  (TR: LF area, HF area) = 0.034,  $P$  (CT: LF area, HF area) = 0.92,  $P$  (LF area: TR, CT) =  $2.44e-08$ ,  $P$  (HF area: TR, CT) =  $2.28e-06$ . (K) Distribution of the monotonicity index (MI) of TR and CT neurons. (L) Comparison of the fraction of non-monotonic neurons of TR and CT.  $N = 23$  focal planes from 5 mice for TR,  $N = 40$  focal planes from 10 mice for CT; "TR":  $20.00\% \backslash 12.92-31.49\%$ , "CT":  $25.00\% \backslash 14.58-40.00\%$ ;  $P = 0.33$ . (M) Comparison of the fraction of non-monotonic neurons of TR and CT in LF and HF areas. LF area:  $N = 4$  focal planes from 3 mice for TR;  $N = 9$  focal planes from 5 mice for CT. HF area:  $N = 5$  focal planes from 5 mice for TR;  $N = 11$  focal planes from 10 mice for CT. "TR: LF area":  $9.55\% \backslash 6.08-13.50\%$ , "TR: HF area":  $15.38\% \backslash 14.66-29.38\%$ , "CT: LF area":  $22.22\% \backslash 17.59-31.67\%$ , "CT: HF area":  $38.18\% \backslash 16.67-60.00\%$ ;  $P$  (TR: LF area, HF area) = 0.032,  $P$  (CT: LF area, HF area) = 0.19,  $P$  (LF area: TR, CT) = 0.016,  $P$  (HF area: TR, CT) = 0.17. (N) Comparison of the MI of non-monotonic and monotonic neurons in TR and CT. Non-mon:  $N = 107$  neurons for TR,  $N = 128$  neurons for CT; "TR":  $0.42 \backslash 0.33-0.47$ , "CT":  $0.35 \backslash 0.26-0.44$ ;  $P = 7.03e-04$ . Mon:  $N = 374$  neurons for TR,  $N = 363$  neurons for CT; "TR":  $0.78 \backslash 0.63-1.00$ , "CT":  $1.00 \backslash 0.73-1.00$ ;  $P = 6.19e-11$ . (O) Comparison of the BF between monotonic and non-monotonic neurons in TR and CT. TR:  $N = 107$  neurons for non-mon,  $N = 374$  neurons for mon; "non-mon":  $6.60 \backslash 2.93-12.10$  kHz, "mon":  $8.90 \backslash 3.60-16.30$  kHz;  $P = 0.69$ . CT:  $N = 128$  neurons for non-mon,  $N = 363$  neurons for mon; "non-mon":  $8.90 \backslash 3.60-19.15$  kHz, "mon":  $6.60 \backslash 2.70-12.10$  kHz;  $P = 1.64e-04$ . (P) Comparison of the  $BW_{max}$  between monotonic and non-monotonic neurons in TR and CT. TR:  $N = 107$  neurons for non-mon,  $N = 374$  neurons for mon; "non-mon":  $1.72 \backslash 0.86-2.15$  octaves, "mon":  $1.29 \backslash 0.86-2.15$  octaves;  $P = 0.92$ . CT:  $N = 128$  neurons for non-mon,  $N = 363$  neurons for mon; "non-mon":  $3.44 \backslash 2.15-3.87$  octaves, "mon":  $3.01 \backslash 2.15-3.87$  octaves;  $P = 0.052$ .

values indicate sharper tuning of neurons, which suggests higher frequency discrimination (Micheyl et al., 2013 [\[1\]](#)), these results suggest that TR neurons might be responsible for finer tone discrimination.

As the bandwidth of neurons' FRA can also reflect the selectivity of overall tonal responses (Rodrigues-Dagaeff et al., 1989 [\[2\]](#); Schreiner and Sutter, 1992 [\[3\]](#)), we examined the maximum FRA width of TR or CT neurons (**Figure 4G-J** [\[4\]](#)).  $BW_{max}$  was defined as the maximum FRA width at any sound level. The quantitative analysis showed that the  $BW_{max}$  of TR neurons was significantly narrower than that of CT neurons, irrespective of the FRA shape (**Figure 4G** [\[5\]](#)). In addition, the distribution of TR neurons generally peaked at a  $BW_{max}$  of 1–2 octaves, while the distribution of CT neurons peaked at a  $BW_{max}$  of 3–4 octaves (**Figure 4H** [\[6\]](#)). The statistical analysis showed that CT neurons had significantly higher  $BW_{max}$  than TR neurons (**Figure 4I** [\[7\]](#)). Within LF or HF areas, TR neurons had a significantly narrower  $BW_{max}$  range than CT neurons in either area. Surprisingly, TR neurons in HF areas had a significantly lower  $BW_{max}$  than TR neurons in LF areas (**Figure 4J** [\[8\]](#)), suggesting that the  $BW_{max}$  distribution among TR neurons might change along the caudal-to-rostral axis.

We then investigated the organization for intensity tuning, another crucial feature in auditory perception, based on the monotonicity index (MI, see Methods for details), wherein 1 indicated monotonic response increase with intensity and 0 indicated strong nonmonotonic tuning (de la Rocha et al., 2008 [\[9\]](#); Sutter and Schreiner, 1995 [\[3\]](#); Watkins and Barbour, 2011 [\[10\]](#)). The MI distribution across both TR and CT neuronal populations peaked at 1 (**Figure 4K** [\[11\]](#)). Subsequent categorization as either monotonic or non-monotonic (using a criterion of  $MI = 0.5$  (de la Rocha et al., 2008 [\[9\]](#); Moore and Wehr, 2013 [\[12\]](#)), neurons with an index  $< 0.5$  considered intensity-tuned), showed that the fraction of neurons exhibiting non-monotonic intensity response did not significantly differ between TR and CT neurons (**Figure 4L** [\[13\]](#)). However, examination of MI values within LF or HF areas showed that a higher proportion of non-monotonic CT neurons than TR were located in LF areas, while a larger proportion of non-monotonic TR neurons were located in HF areas than in LF areas (**Figure 4M** [\[14\]](#)). In addition, among non-monotonic neurons, TR neurons had higher MI than CT neurons, whereas monotonic TR neurons had a lower MI than monotonic CT neurons (**Figure 4N** [\[15\]](#)). Furthermore, we note that non-monotonic CT neurons had a higher BF than monotonic CT neurons, whereas the BF was similar between monotonic and non-monotonic TR neurons (**Figure 4O** [\[16\]](#)). Finally,  $BW_{max}$  did not significantly differ between monotonic and non-monotonic neurons in either the TR or CT populations (**Figure 4P** [\[17\]](#)).

## Discussion

A key finding in this study is that the information flow descending from A1 to the thalamus via CT neurons does not preserve tonotopy, which contrasted with the observation that the information flow ascending from the thalamus to A1 via TR neurons exhibits clear tonotopy.

### Layer specific versus projection specific functional organization in A1

Previous studies have reported the existence of tonotopic gradients across all layers of A1 including L6 (Guo et al., 2012 [\[18\]](#); Tischbirek et al., 2019 [\[19\]](#)). CT neurons labeled with CVS-GCaMP6s were restricted to A1 L6 but did not preserve tonotopy. This is because only 30 - 50% of the pyramidal cells in L6 are CT neurons (Thomson, 2010 [\[20\]](#)), and the neurons we image are restricted to MGB-projecting CT neurons, excluding CT neurons projecting to other downstream nuclei (Clayton et al., 2021 [\[21\]](#)). Using CVS-based labeling approach and 2P single-cell-resolved imaging, we successfully dissected a small fraction of CT neurons from all L6 neurons and studied the functional organization of this group of neurons.

The studies examining functional organization in AuC have mostly focused on layer specificity (Bandyopadhyay et al., 2010 [↗](#); Montes-Lourido et al., 2021 [↗](#); Rothschild et al., 2010 [↗](#); Tischbirek et al., 2019 [↗](#); Winkowski and Kanold, 2013 [↗](#)), and these studies have provided us with a basic understanding of the tonotopy and heterogeneity of AuC. However, layer-specific representation is a relatively coarse approach. Projection specificity-based approach provides detailed insight into differences in population activity and frequency response-related topographic organization between neurons with different connectivity specificity. For example, anatomical studies of AuC in several species have identified the topographical organization of L5 corticocollicular projections (Bajo and Moore, 2005 [↗](#); Saldana et al., 1996 [↗](#); Stebbings et al., 2014 [↗](#)). By contrast, a recent study using a pathway-specific labeling method showed that L5 corticocollicular neurons displayed a relatively weaker topological organization than the non-corticocollicular neurons in the same layer (Schmitt et al., 2023 [↗](#)). Those studies and our data consistently support the notion that different anatomically organized projections do not necessarily transmit information with the same topological alignment. Therefore, combining pathway-specific labeling and *in vivo* single-cell-resolved functional imaging could reveal unexpected fine-scale details and settle discrepancies arising from results obtained by individual methods alone.

## CVS virus provides an effective means of mapping neuronal function in L6

The CVS-N2c-ΔG rabies virus strain, a recently engineered self-inactivating ΔG rabies virus lacking the polymerase gene, exhibits strong neurotropism and reduced cytotoxicity (Reardon et al., 2016 [↗](#)). Previous studies have revealed that reliable neural activity using the CVS-GCaMP6s virus can be maintained for at least 21 days, as recorded by fiber photometry (Lin et al., 2023 [↗](#)). In acute forebrain slices, CVS-N2c-ΔG-hChR2-YFP reliably elicits action potentials in transfected cortical neurons for at least 28 days after infection (Reardon et al., 2016 [↗](#)). Our recent studies also confirm that CVS-labeled CT neurons exhibit electrophysiological properties that are indistinguishable from those of normal neurons (Gu et al., 2023 [↗](#)). Here, chronic 2P calcium imaging shows that CT neurons retain stable sound response properties for at least 13 days (**Figure 1—figure supplementary 1** [↗](#)). All of these studies suggest that the CVS virus-labeling method provides an effective means of mapping neuronal circuitry and manipulating neuronal activity *in vivo* in the mammalian central nervous system.

## Tonotopy of CT neurons can be observed in wide-field but not 2P imaging

Our 2P imaging results are not contradictory to our wide-field epifluorescence imaging data (**Figure 1** [↗](#)), showing a tonotopy in A1 from the same mice that were later used for 2P imaging experiments. First, the source of the fluorescence signal differs between wide-field imaging and 2P imaging. Wide-field epifluorescence imaging reports all fluorescent proteins from cell bodies and neuropils across layers, whereas 2P imaging only examines fluorescent signals from the cell bodies (Scott et al., 2018 [↗](#)). Furthermore, BF tuning from individual neurons results in a significantly closer match to wide-field mapping before removal of the neuropil contribution, thus confirming that the neuropil produces greater local homogeneity in BFs along with a clearer global tonotopic organization (Romero et al., 2020 [↗](#)). For L6 CT neurons, the contribution of the neuropil (dendrites) is important due to the specificity of their morphological structure: the cell bodies are located in L6, while their corresponding apical dendrites terminate in L4 (32%) or extend to L1 (68%) (Olsen et al., 2012 [↗](#)).

## Profiling of auditory features of “in-and-out” neurons in A1

At the single-neuron level, the tonotopic organization or functional distribution of adjacent neuron clusters in the AuC can be highly heterogeneous in mice (Issa et al., 2014 [↗](#); Tao et al., 2017 [↗](#); Winkowski and Kanold, 2013 [↗](#)). Our findings in this study reveal that TR neurons exhibit more homogeneous functional and spatial distribution (**Figure 3** [↗](#)), which contrasts with some

other single-cell-resolved studies in the mouse AuC (Li et al., 2017 [DOI](#); Rothschild et al., 2010 [DOI](#); Winkowski and Kanold, 2013 [DOI](#)). This discrepancy can also be explained by the pathway specificity, i.e., only a small fraction of neurons in the upper layers of AuC are thalamocortical recipients. It is expected that TR neurons possess high-gradient tonotopy and sharp tuning because feedforward thalamocortical projections are known to possess the same features (Guo et al., 2012 [DOI](#); Kanold et al., 2014 [DOI](#); Winkowski and Kanold, 2013 [DOI](#)). What is unexpected is that the same predictions based on the classical literature of functional mapping of AuC (Guo et al., 2012 [DOI](#); Imig et al., 1977 [DOI](#); Recanzone et al., 1999 [DOI](#); Rothschild et al., 2010 [DOI](#)) do not apply to L6 CT neurons. This fits perfectly with a previous study, which found a transition from precise, homogenous frequency organization in L4 to coarse, diffuse organization in L2/3, suggesting that information flow exhibits diverse selectivity after cortical processing (King et al., 2018 [DOI](#); Winkowski and Kanold, 2013 [DOI](#)).

What could be the functional meaning of this change in, or elimination of, tonotopic organization? We suggest that the descending information flow of A1-MGB projection could potentially support higher cortical functions such as higher-level feedback processing of complex (Homma et al., 2017 [DOI](#); Malmierca et al., 2015 [DOI](#)) and behaviorally meaningful sound (Jeschke et al., 2021 [DOI](#); Ohl et al., 2001 [DOI](#); Wang et al., 2020 [DOI](#); Wang et al., 2022 [DOI](#)). In addition to frequency preference, this study also provides evidence illustrating the effects of cortical processing on response properties. For example,  $BW_{max}$ , which reflects the degree of integration of synaptic inputs from presynaptic neurons (Kratz and Manis, 2015 [DOI](#); Li et al., 2019 [DOI](#); Schreiner and Sutter, 1992 [DOI](#)), is 3.0 octaves in CT neurons, but only 1.3 octaves in TR neurons (**Figure 4I** [DOI](#)), indicative of an increase in synaptic integration during cortical processing (Li et al., 2019 [DOI](#)).

We propose that the lack of tonotopy is an active computation, not a passive degradation. CT neurons likely pool inputs from a wide range of upstream neurons with diverse frequency preferences. This broad synaptic integration, reflected in their wider tuning bandwidth, would actively erase the fine-grained frequency map in favor of creating a different kind of representation (Brewer and Barton, 2016 [DOI](#)). This transformation away from a classic sensory map may be critical for the function of corticothalamic feedback. Instead of relaying “what” frequency was heard, the descending signal from CT neurons may convey more abstract, higher-order information, such as the behavioral relevance of a sound, predictions about upcoming sounds, or motor-related efference copy signals that are not inherently frequency-specific (Wang et al., 2020 [DOI](#); Wang et al., 2022 [DOI](#)). The descending A1-to-MGB pathway is often considered to be modulatory, shaping thalamic responses rather than driving them directly. A modulatory signal designed to globally adjust thalamic gain or selectivity may not require, and may even be hindered by, a fine-grained topographical organization.

## Conclusion

In summary, our findings fill in a knowledge gap in the auditory physiology: whereas the corticothalamic feedback projection is known to contribute to higher cognitive processing of behaviorally relevant complex sounds, the basic pure-tone map that underlies and facilitates this advanced processing remained unclear. Our results reveal that the sensory information flow descending from the A1 to the thalamus via L6 CT neurons does not arrange tonotopically. The new shift, categorizing cortical neurons based on their projection specificity, represents an advance in the conceptual framework for functional organization.

## Materials and methods



## Key resources table

| Reagent<br>(species) or resource                        | type | Designation                                           | Source<br>reference     | or<br>Identifiers        | Additional<br>information                                                 |
|---------------------------------------------------------|------|-------------------------------------------------------|-------------------------|--------------------------|---------------------------------------------------------------------------|
| <b>Genetic reagent</b><br><b>(Mus musculus)</b>         |      | Mouse: C57BL/6J                                       | Beijing HFK             | RRID:IMSR_JA<br>X:000664 | <a href="https://www.jax.org/">https://www.jax.org/</a>                   |
| <b>DNA</b><br><b>reagent</b>                            |      | AAV2/1-hSyn-Cre-WP<br>RE-hGH polyA                    | BrainVTA                | PT-0136                  |                                                                           |
| <b>DNA</b><br><b>reagent</b>                            |      | AAV2/9-CaMKII-DIO-<br>GCaMP6s-WPRE-hGH<br>pA          | BrainVTA                | PT-0071                  |                                                                           |
| <b>RNA</b><br><b>reagent</b>                            |      | CVS-N2c-ΔG-GCaM6s                                     | BrainCase Co.,<br>Ltd.  | BC-RV-CVS715             |                                                                           |
| <b>Peptide,</b><br><b>recombinant</b><br><b>protein</b> |      | Alexa<br>555-conjugated<br>Cholera<br>Toxin Subunit B | Fluor<br>Invitrogen     | C34776                   |                                                                           |
| <b>Chemical</b><br><b>compound</b>                      |      | DAPI                                                  | Sigma-Aldrich           | Cat# D9542               |                                                                           |
| <b>Software,</b><br><b>algorithm</b>                    |      | Matlab                                                | Mathworks               | N/A                      | <a href="https://www.mathworks.com/">https://www.mathworks.com/</a>       |
| <b>Software,</b><br><b>algorithm</b>                    |      | LabVIEW                                               | National<br>Instruments | N/A                      | <a href="https://www.ni.com/zh-cn.html">https://www.ni.com/zh-cn.html</a> |
| <b>Software,</b><br><b>algorithm</b>                    |      | Igor Pro                                              | Wavemetrics             | N/A                      | <a href="https://www.wavemetrics.com/">https://www.wavemetrics.com/</a>   |

|                        |                                              |            |     |                           |
|------------------------|----------------------------------------------|------------|-----|---------------------------|
|                        |                                              |            |     | ics.com/                  |
| Software,<br>algorithm | Prism                                        | GraphPad   | N/A | https://www.graphpad.com/ |
| Other                  | UHU Super Glue                               | DETAI      | N/A |                           |
| Other                  | Sun Medical Super Bond C & B Kit Bonding Kit | Super-Bond | N/A |                           |

### Animals

C57BL/6J male mice (2–3 months old) were provided by the Laboratory Animal Center at the Third Military Medical University. The mice were housed in a temperature- and humidity-controlled room on a cycle of 12-h light/dark (lights off at 19:00) with free access to food and water. All animal procedures were approved by the Animal Care Committee of the Third Military Medical University and were performed in accordance with the principles outlined in the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

### Virus injections and confocal imaging

Mice were anesthetized with 1% to 2% isoflurane in pure oxygen and placed in a stereotactic frame (Beijing Zhongshi Dichuang Technology Development Co., Ltd.). A warm heating pad was used to keep the animals at a proper body temperature (36.5°C to 37.5°C). To achieve TR neurons labeling, a Cre-expressing AAV (AAV2/1-hSyn-Cre-WPRE-hGH polyA, titer  $\geq 1\text{E}+13\text{vg/ml}$ , PT-0136, BrainVTA) was injected into the MGB (AP -3.1 mm, ML 2.0 mm, DV -2.8 mm from dura) and DIO-expressing AAV (AAV2/9-CaMKII-DIO-GCaMp6s-WPRE-hGH pA, PT-0071, BrainVTA) was injected into the AuC (AP -3.1 mm, ML 3.8 mm, DV -1.4 mm from dura with manipulator tilted 25° from the z-axis). For CT neuron labeling, the CVS virus (CVS-N2c-ΔG-GCaM6s), packaged by BrainCase Co., Ltd., Shenzhen, China, was injected into the MGB. In all the injections, approximately 100 nL of virus was performed. After injection, the pipette was held in place for 10 min before retraction. Tissue glue (3M Animal Care Products, Vetbond) was used for bonding the scalp incision.

Mice were killed for histology 4 weeks after AAV injection or 7 days after CVS injection. For slice preparation, mice were perfused transcardially with 4% paraformaldehyde in phosphate-buffered saline. Brains were postfixed in 4% paraformaldehyde overnight at 4 °C and cut into 50 μm sections on a cryostat microtome (Thermo Fisher, NX50, Waltham, MA). Mounted sections were imaged on a scanning confocal microscope (TCS SP5, Leica).

### Auditory stimulation

Sound stimuli were delivered by an ED1 electrostatic speaker driver and a free-field ES1 speaker (both from Tucker Davis Technologies). During experiments, the speaker was put at a distance of ~6 cm to the left ear of the animal. The sound stimulus was produced by a custom-written, LabVIEW-based program (LabVIEW 2012, National Instruments) and transformed to analogue voltage through a PCI6731 card (National Instruments). All the data were obtained at 1 MHz via a data acquisition device (USB-6361, National Instruments) and examined by our custom-made

LabVIEW program. For BBN (BBN, bandwidth 0–50 kHz), the sound level was ~65 dB sound pressure level (SPL). We generated a waveform segment of BBN and used the same waveform segment for all experiments involving BBN.

For functional mapping of TR and CT neurons, sequences of randomly ordered pure tones with 11 frequencies (2–40 kHz) at 6 attenuation levels (30–80 dB SPL) were used. Each frequency-attenuation combination was presented 5–8 times. Background noise generated by the recording hardware, especially the resonance scanner of the 2P microscope, was measured to be below < 30 dB SPL for the relevant frequency range. The use of a specialized sound-proof enclosure with wedge-shaped acoustic foam was implemented to significantly reduce external noise interference. These strategies ensured that auditory stimuli were delivered under highly controlled, low-noise conditions, thereby enhancing the reliability and accuracy of the neural response measurements obtained throughout the study. As described in our previous reports, low frequencies (<1 kHz) were major components of background noise. With a spectral density of ~33 dB/sqrt (Hz), the peak of background noise was below 1 kHz. Neither visible light nor other sensory stimuli were present. The duration of a sound stimulus (tone or BBN) was 100 ms.

### ***In vivo* wide-field epifluorescence imaging**

A homemade binocular microscope (BM01, SIBET, CAS) with a 4X, 0.2 NA objective (Olympus) was used to record wide-field fluorescence images in the mouse cortex for establishing the reference cortical map of A1. A light-emitting diode (470 nm, M470L4, Thorlabs) was used for blue illumination. Green fluorescence passed through a filter cube was measured at 10 Hz with a sCMOS camera (Zyla 4.2, Andor Technology). Mice injected with AAV-GCaM6s or CVS-GCaM6s were used to functionally identify the region of A1. The mouse was anesthetized by isoflurane and kept on a warm plate (37.5°C). A piece of bone (~5 mm × 5 mm) was removed and replaced by a coverslip (3 mm in diameter). To localize A1, two pure tones (4 and 32 kHz) were repeatedly presented 20 times at an interval of 6 s.

We observed that pure tone stimulation evoked wide-field signals in only one auditory region in CVS-GCaM6s mice. To confirm the region, we used the criterion that the ventral part of the medial geniculate body (MGBv) is connecting with A1. Injections were performed under visual guidance using 2P excitation. In the experiment, we inserted the electrode into the cortical region at a depth of ~500 μm below the surface. We used Alexa Fluor 555-conjugated cholera toxin subunit B (CTB) as the neural tracer and injected the fluorescent CTB solution with 0.5% in phosphate buffer by pressure (700 mbar) for 15 min. Seven days after the fluorescent CTB injection, the mice were anesthetized with pentobarbital (1.0 g/kg ip). The brain was first dissected out and then it was immersed with 4% paraformaldehyde overnight. To visualize fluorescent tracers, a consecutive series of coronal or horizontal sections (50 μm thick) were prepared using a sliding cryotome, and then all sections were mounted onto glass slides and imaged with a stereoscope (Olympus).

### **Two-photon $\text{Ca}^{2+}$ imaging in A1**

For 2P imaging in head-fixed awake mice, we removed the skin and muscles over the right A1 after local lidocaine injection under isoflurane anesthesia (1–2%). A custom-made plastic chamber (head post) designed for head-fixed mouse experiments was then glued to the skull with cyanoacrylic glue (UHU). After recovery from surgery for 3 days, the mouse underwent head-fixation training for 3–5 days (from 1 to 4 h per day). After head-fixation training, mice gradually adapted to this posture and were able to sit comfortably for 4 h. On the recording day, a small craniotomy was performed under local anesthesia. Then the field of interest was determined by comparing the wide-field map with the blood vessel patterns to ensure A1 was imaged.

Two-photon imaging was performed with a custom-built 2P microscope system based on a 12.0 kHz resonant scanner (model “LotosScan 1.0”, Suzhou Institute of Biomedical Engineering and Technology). Two-photon excitation light was delivered by a mode-locked Ti: Sa laser (model “Mai-

Tai DeepSee”, Spectra Physics), and a 40×/0.8 numerical aperture (NA) water immersion objective (Nikon) was used for imaging. For  $\text{Ca}^{2+}$  imaging experiments, the excitation wavelength was set to 920 nm. The typical size of the field of view (FOV) was  $\sim 200 \mu\text{m} \times 200 \mu\text{m}$ . The average power delivered to the brain was in the range of 30–120 mW, depending on the depth of imaging.

## Chronic 2P $\text{Ca}^{2+}$ imaging in L6

For chronic 2P imaging, A circular cranial window (2.5 mm diameter coverslip) was implanted over the right AuC after head-post implantation. To this end, the muscle and skull fascia above the temporal skull were removed after local lidocaine (2%) injection. Afterward, a custom-made plastic chamber was fixed to the skull with dental cement (Superbond, Sun Medical Co., Ltd.) and a small craniotomy ( $\sim 2.7$  mm in diameter, the center point: AP: -3.0 mm, ML: -4.5 mm) was performed. The dura was removed, and the craniotomy was polished to match the size of the coverslip. A coverslip was carefully placed on top of the cortex with mild compression by tweezers. The coverslip was sealed with UV-curing dental cement (Tetric N-flow, Ivoclar Co., Ltd.). Antibiotics (Cefazolin, 500 mg/kg, North China Pharmaceutical Group Corporation) were administered before surgery, as well as until 3 days after surgery. 3 days of post-surgery recovery were needed before head-fixation training.

Continuous 2P  $\text{Ca}^{2+}$  imaging was performed on day 7 and 13. To minimize the bleaching of cells, we used low laser illumination in the deep imaging, based on high-quality imaging of CT neurons with CVS labeling strategy. The repeated imaging FOVs were identified on consecutive days based on superficial blood vessels and nearby blood vessels, then further refined by visually matching reference images acquired from precedent days.

## Quantification and statistical analysis

Data were analyzed using custom-written software in LabVIEW 2012 (National Instruments), Igor Pro 5.0 (Wavemetrics), Image 1.51 (NIH), Prism 8.4 (GraphPad) and MATLAB 2014a (MathWorks).

## Wide-field imaging data analysis

In each mouse, the recorded cortical images were first down sampled from the original  $750 \times 1200$  pixels to  $75 \times 120$  pixels. After that, the frames recorded with sound stimuli were averaged across 20 trials. To enhance the signal-to-noise ratio, spatial averaging was conducted over  $5 \times 5$  pixels by a matrix filter, and temporal averaging was conducted with three consecutive images. The pre-processed images were then temporally normalized to obtain the relative changes in fluorescence ( $f$ ) pixel-by-pixel. With the baseline fluorescence ( $f_0$ ) obtained by averaging the images of 800 ms before sound stimulation, the relative fluorescence changes of each pixel were calculated as  $\Delta f/f = (f - f_0)/f_0$ . The normalized images are shown on a color-coded scale to visualize the relative fluorescence changes ( $\Delta f/f$ ) in the cortex.

## Two-photon imaging data analysis

To correct motion-related artifacts in imaging data, a frame-by-frame alignment algorithm was used to minimize the sum of squared intensity differences between each frame image and a template, which was the average of the selected image frames. To extract fluorescence signals, neurons were visually identified, and drawing regions of interest (ROIs) based on fluorescence intensity was performed. Fluorescence changes ( $f$ ) were calculated by averaging the corresponding pixel values for each ROI. Relative fluorescence changes  $\Delta f/f = (f - f_0)/f_0$  were calculated as  $\text{Ca}^{2+}$  signals, where the baseline fluorescence  $f_0$  was estimated as the 25th percentile of the entire fluorescence recording. To calculate the amplitude of sound-evoked  $\text{Ca}^{2+}$  transients, we performed automatic  $\text{Ca}^{2+}$  transient detection based on threshold criteria regarding peak amplitude and rising rate. The noise level was set to be 3 times the standard deviation of the baseline (window length: 1 s). The peak amplitude and the rate of rising of the  $\text{Ca}^{2+}$  signals were

calculated to determine whether it was a true transient. The trace of the detected  $\text{Ca}^{2+}$  transient was first extracted by exponential infinite impulse response (IIR) filtering (window length: 200 ms) and then subtracted from the original signal. The residual fluorescence trace was used as the baseline for the next transient detection, similar to previously published peeling approaches.

## Sound-evoked $\text{Ca}^{2+}$ responsiveness

### BBN-evoked $\text{Ca}^{2+}$ responsiveness

For data from chronic  $\text{Ca}^{2+}$  imaging, we tracked the same FOV based on the last training day. We removed the outer ~10% of the image from each ROI to account for edge effects or imaging deviation. The success rate was defined as the fraction of sound-evoked responses during 10 consecutive BBN stimuli. Note that “Active” neurons showed clear  $\text{Ca}^{2+}$  transients during the entire recording duration, including spontaneous and BBN-evoked responses. The neurons with a success rate  $\geq 50\%$  were defined as “Reliable” neurons among active neurons.

### Pure tone-evoked $\text{Ca}^{2+}$ responsiveness

The frequency response area (FRA) of each ROI was constructed from the 55 average responses to all of the unique frequency-attenuation combinations. The frequency tuning curves were constructed by plotting the average values (and the s.e.m.) of the calcium signal amplitudes from single trials for each frequency tested. If more than one area of contiguous frequency-level combinations remained, the largest one was defined as the FRA. The amplitude of a  $\text{Ca}^{2+}$  signal was determined as the average value for a period of 2001ms around the peak of the calcium transient. The baseline value was calculated for a period of 1001ms before the onset of the auditory stimulus. Pure tone frequencies that induced response amplitudes higher than half of the maximal response were defined as effective frequencies.

“Irregular” neurons were characterized by exhibiting spontaneous activity patterns that were highly variable and inconsistent in their responses to sound stimulation. These neurons showed no clear or predictable firing pattern. “Tuned” neurons represented a subset of responsive neurons that demonstrated significant and consistent selectivity for specific auditory stimuli. These neurons exhibited well-defined frequency tuning or preference, responding robustly to certain sound features while showing diminished activity to others. “Silent” neurons were operationally defined as those that remained completely inactive throughout the entire recording period, which extended beyond 30 minutes. These neurons showed no detectable spontaneous firing or evoked responses during the experiments. For tuned neurons, the best frequency (BF) was defined as the sound frequency associated with the highest response averaged across all sound levels.

FRAs that exhibited a pattern of decreasing frequency selectivity as sound intensity increased were categorized as V-shaped, reflecting a broadening of the receptive field at higher stimulus levels. This shape indicates that the neuron responds to a wider range of frequencies when the stimulus becomes more intense, suggesting a loss of tuning precision at higher intensities. In contrast, FRAs that maintained consistent frequency selectivity across increasing sound intensities were classified as I-shaped, signifying a stable tuning profile regardless of stimulus amplitude. This pattern implies that the neuron’s frequency preference remains sharply defined and resistant to intensity-dependent modulation. Additionally, FRAs that demonstrated responsiveness confined to a narrow range of both sound frequencies and intensities were designated as O-shaped, provided that their peak neural response did not occur at the maximum intensity tested. This classification suggests a limited dynamic range and a preference for intermediate stimulus conditions. Maximum bandwidth ( $\text{BW}_{\text{max}}$ ) was defined as the maximum FRA width at any level.



To compare the BF heterogeneity, we analyzed the BF distribution in fixed-size analysis windows. The BF heterogeneity was quantified by the interquartile range (IQR) of the distribution. For field IQR, if less than 5 neurons were within a 100  $\mu\text{m}$  radius (including the center neuron), no IQR was calculated (Schmitt et al., 2023 [DOI](#)). For Large-scale analysis, imaging regions in separate mice were aligned with each other according to the caudal-to-rostral gradient that was identifiable in the wide-field images.

The monotonicity index (MI) characterizes the strength of a neuron's intensity tuning. We defined MI as the neuron's response at maximum intensity (80 dB) divided by its maximum response (Moore and Wehr, 2013 [DOI](#); Sutter and Schreiner, 1995 [DOI](#)). An MI of 1 indicates no intensity tuning; an MI near zero indicates very strong intensity tuning.

## Statistics

To compare data between groups, we used the nonparametric Wilcoxon rank sum test (unpaired), and Wilcoxon signed-rank test (paired) to determine statistical significance ( $P < 0.05$ ) between them. In the text, summarized data are presented as the median\25th–75th percentiles. In the figures, the data presented in the box-and-whisker plot indicate the median (center line), 25th and 75th percentiles (Q1 and Q3), i.e., IQR (box),  $Q1 - 1.5 \times \text{IQR}$  and  $Q3 + 1.5 \times \text{IQR}$  (whiskers), and all other data with error bars are presented as the mean  $\pm$  s.e.m.

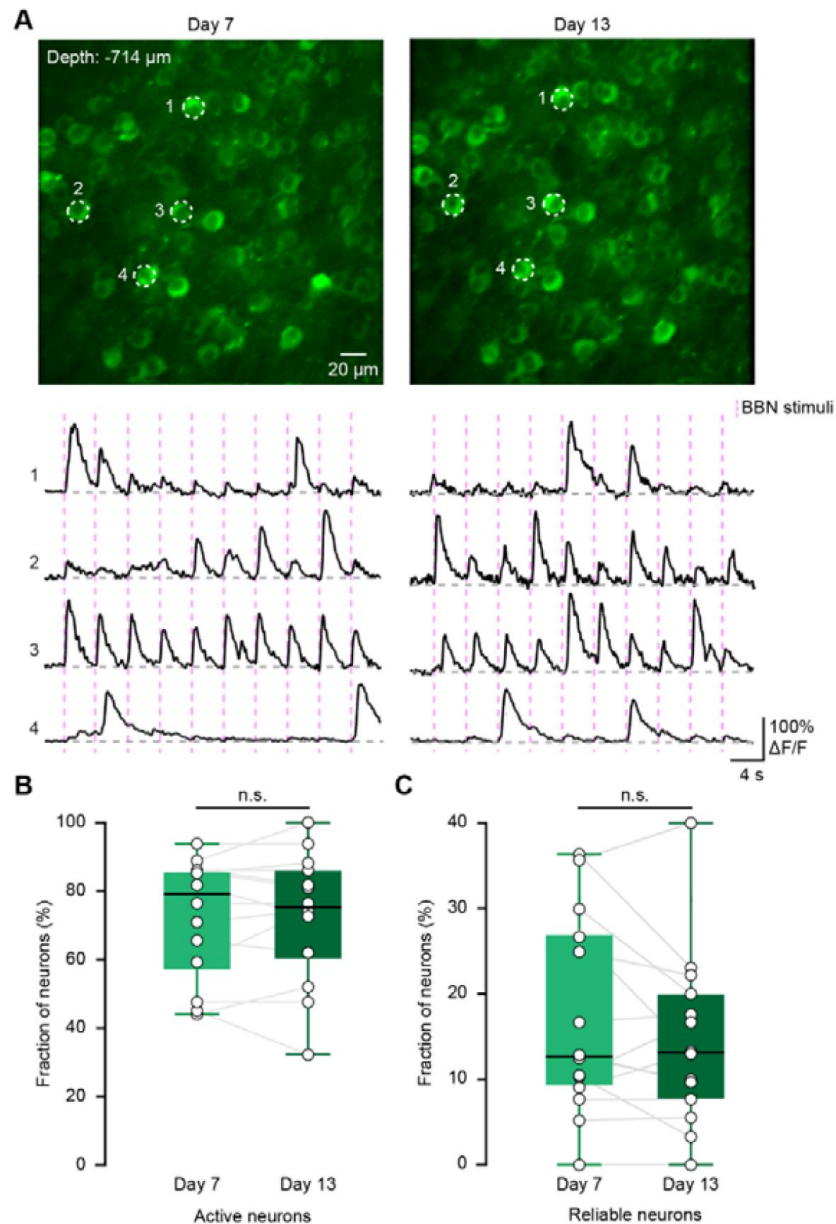
## Data availability

Wide-field imaging and two-photon imaging are available upon request from the corresponding author. Source data underlying **Figs. 2** [DOI](#)–**4** [DOI](#) is available as a Source data file. No datasets that require mandatory deposition into a public database were generated during the current study. Source data are provided in this paper.

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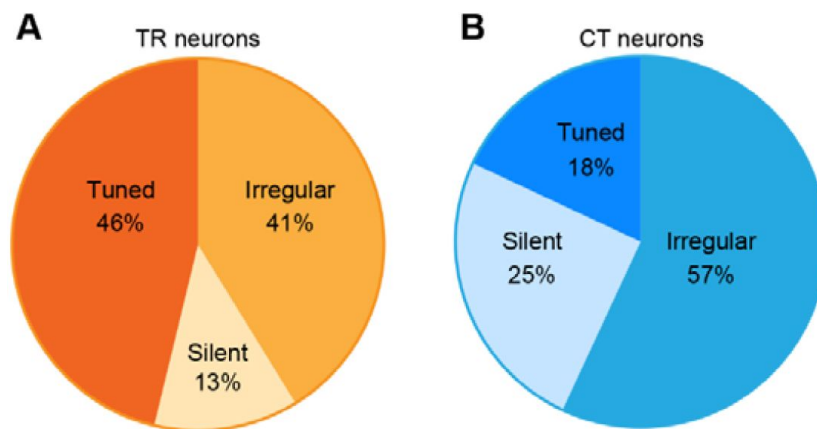
## Additional information



**Figure 1- figure supplementary 1.**

### Chronic *in vivo* 2P imaging of CT neurons.

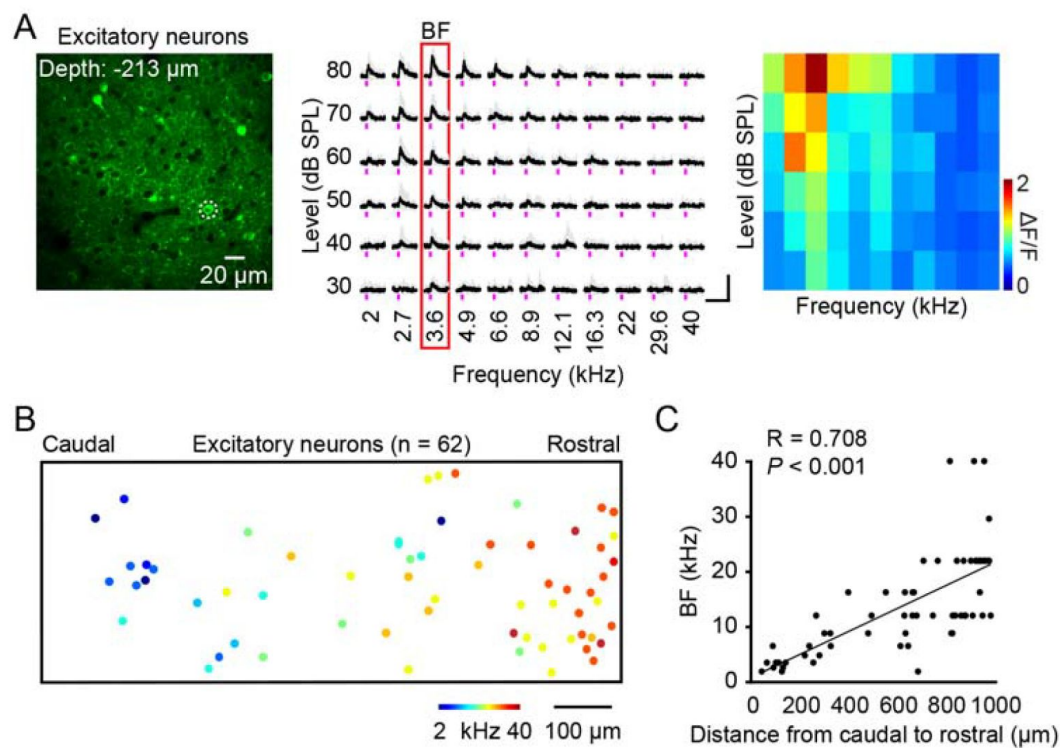
(A) Calcium imaging from L6 CT neurons in a CVS-expressing mouse, again with repeated imaging of the same focal planes on day 7 and 13 after virus injection. The dashed circle indicates the example neurons, showing fluorescence traces of neurons in the lower panel. (B) The comparison of the fraction of active CT neurons on day 7 and 13.  $N = 14$  focal planes from 5 mice. "Day 7": 79.14\59.26–86.11%, "Day 13": 75.33\62.07–86.11%;  $P = 0.96$ , two-sided Wilcoxon signed-rank test,  $*P < 0.05$ ,  $**P < 0.01$ , and  $***P < 0.001$ , same statistics for boxplots. (C) The comparison of the fraction of reliable CT neurons on day 7 and 13. "Day 7": 12.70\9.09–26.67%, "Day 13": 13.19\7.69–20.00%;  $P = 0.53$ . The definitions of "active" and "reliable" neurons are provided in METHOD DETAILS.



**Figure 2- figure supplementary 2.**

**Pie charts showing the percentage of responsive neurons to pure-tone stimulation of TR and CT neurons.**

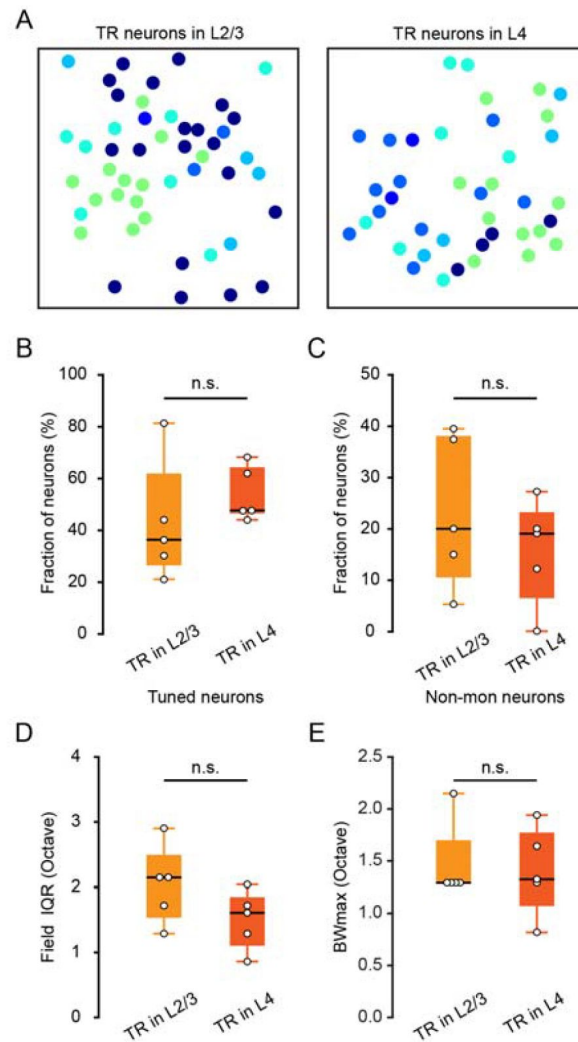
*N* = 5 animals for TR neurons, *N* = 10 animals for CT neurons. The definitions of “silent” and “irregular” neurons are seen in METHOD DETAILS.



**Figure 2- figure supplementary 3.**

### **Tonotopic gradients of general excitatory neurons in superficial layers.**

(A) Left panel: two-photon imaging of excitatory neurons using AAV-CaMKII-GCaMP6s labeling. The dashed circle indicates the tuned neurons. Fluorescence traces of neurons in the middle panel are indicated by the dashed circle ordered according to sound frequency and level. Middle panel: the five traces associated with the five repeats of each stimulus are plotted in gray. The average calcium signals are plotted in black. The red outline marks the BF response of the neuron. Right panel: the FRA from the tuned neurons in the left panel dashed circle. (B) Reconstruction of the relative spatial locations of tuned general excitatory neurons that covered the whole A1 area and are color-coded according to each neuron's BF. N = 62 neurons from one mouse. (C) Plots of excitatory neurons' BF and their relative distances along the caudal-to-rostral axis.

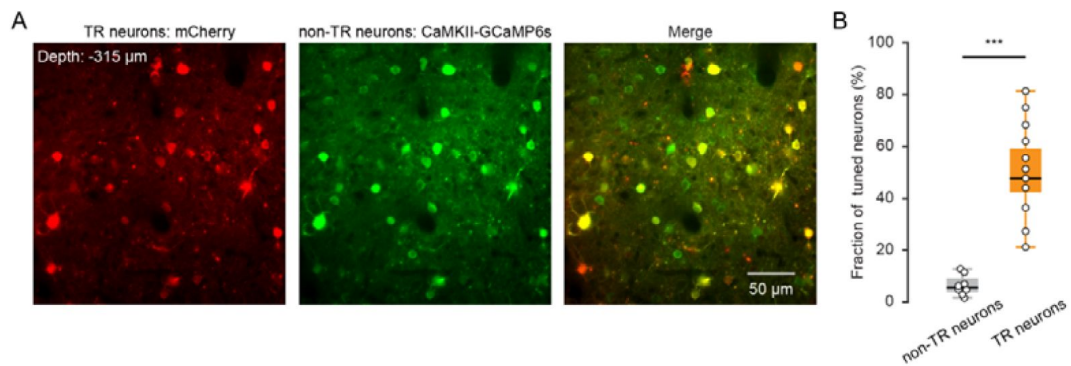


**Figure 2- figure supplementary 4.**

### Comparison of TR neurons across superficial layers.

(**A**) Spatial distribution of all tone-selective neurons in an example imaging plane of TR neurons in L2/3 (left) and L4 (right) imaging area. (**B**) Comparison of the fraction of tuned TR neurons in L2/3 and L4. (**C**) Comparison of the fraction of non-monotonic TR neurons in L2/3 and L4. (**D**) Comparison of the field (200  $\mu$ m) IQR of TR neurons in L2/3 and L4. (**E**) Comparison of the BWmax of TR neurons in L2/3 and L4. N = 5 focal planes for TR neurons in L2/3, N = 5 focal planes for TR neurons in L4.





**Figure 2- figure supplementary 5.**

### Imaging pure tone responses of non-TR neurons.

(**A**) Left panel: example of image with mCherry-expression in TR neurons. Injection site of AAV2/1-Cre in the MGB and AAV2/9-hSyn-DIO-mCherry in A1. Middle panel: example of image with GCaMP6s-expression in general excitatory neurons. Injection site of AAV2/9-CaMKII-GCaMP6s in A1. Right panel: merged image of left panel and middle panel. (**B**) Comparison of the fraction of tuned neurons among non-TR and TR neurons. non-TR neurons: N = 8 local planes from 4 mice. TR neurons: N = 11 local planes from 5 mice.

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## Author contributions

X.C. and H.J. conceived the project. X.C., JX.Z. and Y.Z. designed the experiments. M.G. performed the experiments with the help of R.L., K.L., X.W. and JH.Z.; X.C., H.J., Y.Z., S.L., JX.Z., and M.G. devised the data analysis methods; M.G., S.L., Yun.Z., X.L., C.Z. and JX.Z. performed the data analysis; JX.Z., H.J., Y.Z. and X.C. inspected the data and evaluated the findings; M.G., JX.Z., Y.Z., F.O., X.C. and H.J. wrote the manuscript with the help from all authors.

## Ethics

All experiments were approved by the Animal Care Committee of the Third Military Medical University (Approval number: AMUWEC20230061).

## Code availability

The codes supporting the current study have not been deposited in a public repository, but are available from the corresponding author upon request.

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## **Reviewer #1 (Public review):**

Summary:

In this study, Gu et al., employed novel viral strategies, combined with in vivo two-photon imaging, to map the tone response properties of two groups of cortical neurons in A1 - The thalamocortical recipient (TR neurons) and the corticothalamic (CT neurons). They observed a clear tonotopic gradient among TR neurons but not in CT neurons. Moreover, CT neurons exhibited high heterogeneity of their frequency tuning and broader bandwidth, suggesting increased synaptic integration in these neurons. By parsing out different projecting-specific neurons within A1, this study provides insight into how neurons with different connectivity can exhibit different frequency response-related topographic organization.

Strengths:

This study reveals the importance of studying neurons with projection specificity rather than layer specificity since neurons within the same layer have very diverse molecular, morphological, physiological, and connectional features. By utilizing a newly developed rabies virus CSN-N2c GCaMP-expressing vector, the authors can label and image specifically the neurons (CT neurons) in A1 that project to the MGB. To compare, they used an anterograde trans-synaptic tracing strategy to label and image neurons in A1 that receive input from MGB (TR neurons).

Weaknesses:

- Perhaps as cited in the introduction, it is well known that tonotopic gradient is well preserved across all layers within A1, but I feel if the authors want to highlight the specificity of their virus tracing strategy and the populations that they imaged in L2/3 (TR neurons) and L6 (CT neurons), they should perform control groups where they image general excitatory neurons in the two depths and compare to TR and CT neurons, respectively. This will show that it's not their imaging/analysis or behavioral paradigms that are different from other labs.

- Fig 1D and G, the y-axis is Distance from pia (%). I'm not exactly sure what this means. How does % translate to real cortical thickness?

- For Fig. 2G and H, is each circle a neuron or an animal? Why are they staggered on top of each other on the x-axis? If x-axis is the distance from caudal to rostral, each neuron should have a different distance? Also, it seems like it's because Fig. 2H has more circles, that's why it has more variation thus not significant (for example, at 600 or 900um, 2G seems to have less circles than 2H).

- Similar in Fig 2J and L, why are the circles staggered on the y-axis now? And is each circle now a neuron or a trial? It seems they have much more circles than Fig 2G and 2H. Also I don't think doing a correlation is the proper stats for this type of plot (this point applies to Fig. 3H and 3J)

- What does inter-quartile range of BF (IQRBF, in octaves) imply? What's the interpretation of this analysis? I am confused why TR neurons show high IQR in HF areas compared to LF areas mean homogeneity among TR neurons (line 213 - 216). On the same note, how is this different from the BF variability? Isn't higher IQR = to higher variability?

- Fig. 4A-B, there's no clear criteria on how the authors categorize V, I, and O Shape. The descriptions in the Methods (line 721 - 725) are also very vague.

Comments on revisions:

The authors have addressed all my questions in the previous round.

<https://doi.org/10.7554/eLife.99989.2.sa3>

## Reviewer #2 (Public review):

Summary:

Gu and Liang et. al investigated how auditory information is mapped and transformed as it enters and exits a auditory cortex. They use anterograde transsynaptic tracers to label and perform calcium imaging of thalamorecipient neurons in A1 and retrograde tracers to label and perform calcium imaging of corticothalamic output neurons. They demonstrate a degradation of tonotopic organization from the input to output neurons.

Strengths:

The experiments appear well executed, well described, and analyzed.

Weaknesses:

(1) Given that the CT and TR neurons were imaged at different depths, the question as to whether not these differences could otherwise be explained by layer-specific differences is still not 100% resolved. Control measurements would be needed either by recording 1) CT

neurons upper layers 2) TR in deeper layers 3) non-CT in deeper layers and/or 4) non-TR in upper layers.

(2) What percent of the neurons at the depths being are CT neurons? Similar questions for TR neurons?

(3) V-shaped, I-shaped, or O-shaped is not an intuitively understood nomenclature, consider changing. Further, the x/y axis for Figure 4a is not labeled, so it's not clear what the heat maps are supposed to represent.

(4). Many references about projection neurons and cortical circuits are based on studies from visual or somatosensory cortex. Auditory cortex organization is not necessarily the same as other sensory areas. Auditory cortex references should be used specifically, and not sources reporting on S1, V1.

Comments on revisions:

The authors have fully addressed my concerns.

<https://doi.org/10.7554/eLife.99989.2.sa2>

### Reviewer #3 (Public review):

Summary:

The authors performed wide-field and 2-photon imaging in vivo in awake head-fixed mice, to compare receptive fields and tonotopic organization in thalamocortical recipient (TR) neurons vs corticothalamic (CT) neurons of mouse auditory cortex. TR neurons were found in all cortical layers while CT neurons were restricted to layer 6. The TR neurons at nominal depths of 200-400 microns have a remarkable degree of tonotopy (as good if not better than tonotopic maps reported by multiunit recordings). In contrast, CT neurons were very heterogenous in terms of their best frequency (BF), even when focusing on the low vs high frequency regions of primary auditory cortex. CT neurons also had wider tuning.

Strengths:

This is a thorough examination using modern methods, helping to resolve a question in the field with projection-specific mapping.

Weaknesses:

There are some limitations due to the methods, and it's unclear what the importance of these responses are outside of behavioral context or measured at single timepoints given the plasticity, context-dependence, and receptive field 'drift' that can occur in cortex.

(1) Probably the biggest conceptual difficulty I have with the paper is comparing these results to past studies mapping auditory cortex topography, mainly due to differences in methods. Conventionally, tonotopic organization is observed for characteristic frequency maps (not best frequency maps), as tuning precision degrades and best frequency can shift as sound intensity increases. The authors used six attenuation levels (30-80 dB SPL) and report that the background noise of the 2-photon scope is <30 dB SPL, which seems very quiet. The authors should at least describe the sound-proofing they used to get the noise level that low, and some sense of noise across the 2-40 kHz frequency range would be nice as a supplementary figure. It also remains unclear just what the 2-photon dF/F response represents in terms of spikes. Classic mapping using single-unit or multi-unit electrodes might be sensitive to single spikes (as might be emitted at characteristic frequency), but this might not be as obvious for Ca2+ imaging. This isn't a concern for the internal comparison here between TR and CT cells as

conditions are similar, but is a concern for relating the tonotopy or lack thereof reported here to other studies.

(2) It seems a bit peculiar that while 2721 CT neurons (N=10 mice) were imaged, less than half as many TR cells were imaged (n=1041 cells from N=5 mice). I would have expected there to be many more TR neurons even mouse for mouse (normalizing by number of neurons per mouse), but perhaps the authors were just interested in a comparison data set and not being as thorough or complete with the TR imaging?

(3) The authors definitions of neuronal response type in the methods needs more quantitative detail. The authors state: ""Irregular" neurons exhibited spontaneous activity with highly variable responses to sound stimulation. "Tuned" neurons were responsive neurons that demonstrated significant selectivity for certain stimuli. "Silent" neurons were defined as those that remained completely inactive during our recording period (> 30 min). For tuned neurons, the best frequency (BF) was defined as the sound frequency associated with the highest response averaged across all sound levels." The authors need to define what their thresholds are for 'highly variable', 'significant', and 'completely inactive'. Is best frequency the most significant response, the global max (even if another stimulus evokes a very close amplitude response), etc.

Comments on revisions:

I think the authors misunderstood my point about sound level and characteristic frequency vs best frequency tonotopic maps. Yes, many studies of cortical responses present stimuli at higher intensities than the characteristic frequencies, but as tuning curves widen with sound level, the macroscopic tonotopic organization of primary auditory cortex breaks down at higher intensities. This is why most of the classic studies of tonotopy e.g., from the Merzenich lab) generated maps of characteristic frequency. As I mentioned before, this isn't so much of an issue for the authors' comparisons of TR vs CT organization in their own study, but in general, this makes it difficult to compare aspects of spatially-organized tonotopy from imaging studies with the older electrophysiological 'truer' tonotopic maps. That said, this means that CT cells also might be tonotopically organized if the authors had been able to look at lower intensity tuning properties.

<https://doi.org/10.7554/eLife.99989.2.sa1>

### Author response:

The following is the authors' response to the original reviews.

#### **Reviewer #1 (Public Review):**

##### *Summary:*

*In this study, Gu et al. employed novel viral strategies, combined with in vivo two-photon imaging, to map the tone response properties of two groups of cortical neurons in A1. The thalamocortical recipient (TR neurons) and the corticothalamic (CT neurons). They observed a clear tonotopic gradient among TR neurons but not in CT neurons. Moreover, CT neurons exhibited high heterogeneity of their frequency tuning and broader bandwidth, suggesting increased synaptic integration in these neurons. By parsing out different projecting-specific neurons within A1, this study provides insight into how neurons with different connectivity can exhibit different frequency response-related topographic organization.*

##### *Strengths:*

*This study reveals the importance of studying neurons with projection specificity rather than layer specificity since neurons within the same layer have very diverse molecular, morphological, physiological, and connectional features. By utilizing a newly developed rabies virus CSN-N2c GCaMP-expressing vector, the authors can label and image specifically the neurons (CT neurons) in A1 that project to the MGB. To compare, they used an anterograde trans-synaptic tracing strategy to label and image neurons in A1 that receive input from MGB (TR neurons).*

**Weaknesses:**

*Perhaps as cited in the introduction, it is well known that tonotopic gradient is well preserved across all layers within A1, but I feel if the authors want to highlight the specificity of their virus tracing strategy and the populations that they imaged in L2/3 (TR neurons) and L6 (CT neurons), they should perform control groups where they image general excitatory neurons in the two depths and compare to TR and CT neurons, respectively. This will show that it's not their imaging/analysis or behavioral paradigms that are different from other labs.*

We thank the reviewer for these constructive suggestions. As recommended, we have performed control experiments that imaged the general excitatory neurons in superficial layers (shown below), and the results showed a clear tonotopic gradient, which was consistent with previous findings (Bandyopadhyay et al., 2010; Romero et al., 2020; Rothschild et al., 2010; Tischbirek et al., 2019), thereby validating the reliability of our imaging/analysis approach. The results are presented in a new supplemental figure (Figure 2- figure supplementary 3).

**Related publications:**

- (1) Gu M, Li X, Liang S, Zhu J, Sun P, He Y, Yu H, Li R, Zhou Z, Lyu J, Li SC, Budinger E, Zhou Y, Jia H, Zhang J, Chen X. 2023. Rabies virus-based labeling of layer 6 corticothalamic neurons for two-photon imaging in vivo. *iScience* 26: 106625. DIO: <https://doi.org/10.1016/j.isci.2023.106625>, PMID: 37250327
- (2) Bandyopadhyay S, Shamma SA, Kanold PO. 2010. Dichotomy of functional organization in the mouse auditory cortex. *Nat Neurosci* 13: 361-8. DIO: <https://doi.org/10.1038/nn.2490>, PMID: 20118924
- (3) Romero S, Hight AE, Clayton KK, Resnik J, Williamson RS, Hancock KE, Polley DB. 2020. Cellular and Widefield Imaging of Sound Frequency Organization in Primary and Higher Order Fields of the Mouse Auditory Cortex. *Cerebral Cortex* 30: 1603-1622. DIO: <https://doi.org/10.1093/cercor/bhz190>, PMID: 31667491
- (4) Rothschild G, Nelken I, Mizrahi A. 2010. Functional organization and population dynamics in the mouse primary auditory cortex. *Nat Neurosci* 13: 353-60. DIO: <https://doi.org/10.1038/nn.2484>, PMID: 20118927
- (5) Tischbirek CH, Noda T, Tohmi M, Birkner A, Nelken I, Konnerth A. 2019. In Vivo Functional Mapping of a Cortical Column at Single-Neuron Resolution. *Cell Rep* 27: 1319-1326 e5. DIO: <https://doi.org/10.1016/j.celrep.2019.04.007>, PMID: 31042460

*Figures 1D and G, the y-axis is Distance from pia (%). I'm not exactly sure what this means. How does % translate to real cortical thickness?*

We thank the reviewer for this question. The distance of labeled cells from pia was normalized to the entire distance from pia to L6/WM border for each mouse, according to the

previous study (Chang and Kawai, 2018). For all mice tested, the entire distance from pia to L6/WM border was  $826.5 \pm 23.4$  mm (in the range of 752.9 to 886.1).

Related publications:

Chang M, Kawai HD. 2018. A characterization of laminar architecture in mouse primary auditory cortex. *Brain Structure and Function* 223: 4187-4209. DIO: <https://doi.org/10.1007/s00429-018-1744-8>, PMID: 30187193

*For Figure 2G and H, is each circle a neuron or an animal? Why are they staggered on top of each other on the x-axis? If the x-axis is the distance from caudal to rostral, each neuron should have a different distance? Also, it seems like it's because Figure 2H has more circles, which is why it has more variation, thus not significant (for example, at 600 or 900um, 2G seems to have fewer circles than 2H).*

We sincerely appreciate the reviewer's careful attention to the details of our figures. Each circle in the Figure 2G and H represents an individual imaging focal plane from different animals, and the median BF of some focal planes may be similar, leading to partial overlap. In the regions where overlap occurs, the brightness of the circle will be additive.

Since fewer CT neurons, compared to TR neurons, responded to pure tones within each focal plane, as shown in Figure 2- figure supplementary 2, a larger number of focal planes were imaged to ensure a consistent and robust analysis of the pure tone response characteristics. The higher variance and lack of correlation in CT neurons is a key biological finding, not an artifact of sample size. The data clearly show a wide spread of median BFs at any given location for CT neurons, a feature absent in the TR population.

*Similarly, in Figures 2J and L, why are the circles staggered on the y-axis now? And is each circle now a neuron or a trial? It seems they have many more circles than Figure 2G and 2H. Also, I don't think doing a correlation is the proper stats for this type of plot (this point applies to Figures 3H and 3J).*

We regret any confusion have caused. In fact, Figure 2 illustrates the tonotopic gradient of CT and TR neurons at different scales. Specifically, Figures 2E-H present the imaging from the focal plane perspective (23 focal planes in Figures 2G, 40 focal planes in Figures 2H), whereas Figures 2I-L provide a more detailed view at the single-cell level (481 neurons in Figures 2J, 491 neurons in Figures 2L). So, Figures 2J and L do indeed have more circles than Figures 2G and H. The analysis at these varying scales consistently reveals the presence of a tonotopic gradient in TR neurons, whereas such a gradient is absent in CT neurons.

We used Pearson correlation as a standard and direct method to quantify the linear relationship between a neuron's anatomical position and its frequency preference, which is widely used in the field to provide a quantitative measure (R-value) and a significance level (p-value) for the strength of a tonotopic gradient. The same statistical logic applies to testing for spatial gradients in local heterogeneity in Figure 3. We are confident that this is an appropriate and informative statistical approach for these data.

*What does the inter-quartile range of BF (IQRBF, in octaves) imply? What's the interpretation of this analysis? I am confused as to why TR neurons show high IQR in HF areas compared to LF areas, which means homogeneity among TR neurons (lines 213 - 216). On the same note, how is this different from the BF variability? Isn't higher IQR equal to higher variability?*

We thank the reviewer for raising this important point. IQRBF, is a measure of local tuning heterogeneity. It quantifies the diversity of BFs among neighboring neurons. A small IQRBF



means neighbors are similarly tuned (an orderly, homogeneous map), while a large IQRBF means neighbors have very different BFs (a disordered, heterogeneous map). (Winkowski and Kanold, 2013; Zeng et al., 2019).

From the BF position reconstruction of all TR neurons (Figures 2I), most TR neurons respond to high-frequency sounds in the high-frequency (HF) region, but some neurons respond to low frequencies such as 2 kHz, which contributes to high IQR in HF areas. This does not contradict our main conclusion, that the TR neurons is significantly more homogeneous than the CT neurons. BF variability represents the stability of a neuron's BF over time, while IQR represents the variability of BF among different neurons within a certain range. (Chambers et al., 2023).

Related publications:

- (1) Chambers AR, Aschauer DF, Eppler JB, Kaschube M, Rumpel S. 2023. A stable sensory map emerges from a dynamic equilibrium of neurons with unstable tuning properties. *Cerebral Cortex* 33: 5597-5612. DIO: <https://doi.org/10.1093/cercor/bhac445>, PMID: 36418925
- (2) Winkowski DE, Kanold PO. 2013. Laminar transformation of frequency organization in auditory cortex. *Journal of Neuroscience* 33: 1498-508. DIO: <https://doi.org/10.1523/JNEUROSCI.3101-12.2013>, PMID: 23345224
- (3) Zeng HH, Huang JF, Chen M, Wen YQ, Shen ZM, Poo MM. 2019. Local homogeneity of tonotopic organization in the primary auditory cortex of marmosets. *Proceedings of the National Academy of Sciences of the United States of America* 116: 3239-3244. DIO: <https://doi.org/10.1073/pnas.1816653116>, PMID: 30718428

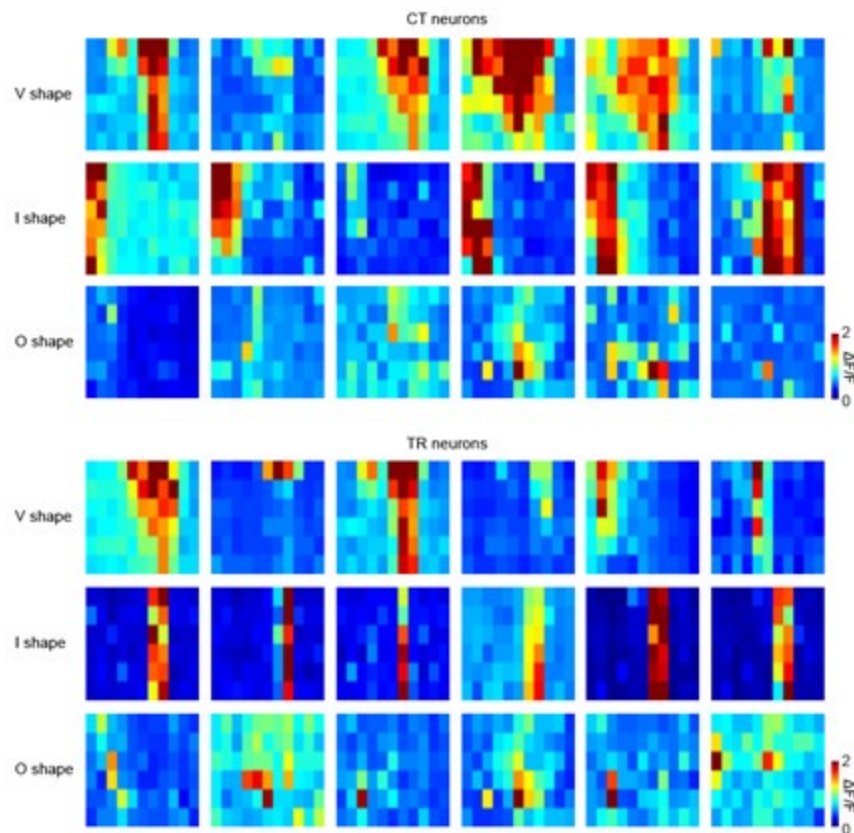
*Figure 4A-B, there are no clear criteria on how the authors categorize V, I, and O shapes. The descriptions in the Methods (lines 721 - 725) are also very vague.*

We apologize for the initial vagueness and have replaced the descriptions in the Methods section. “V-shaped”: Neurons whose FRAs show decreasing frequency selectivity with increasing intensity. “I-shaped”: Neurons whose FRAs show constant frequency selectivity with increasing intensity. “O-shaped”: Neurons responsive to a small range of intensities and frequencies, with the peak response not occurring at the highest intensity level.

To provide better visual intuition, we show multiple representative examples of each FRA type for both TR and CT neurons below. We are confident that these provide the necessary clarity and reproducibility for our analysis of receptive field properties.

#### Author response image 1.

Different FRA types within the dataset of TR and CT neurons. Each row shows 6 representative FRAs from a specific type. Types are V-shaped (‘V’), I-shaped (‘I’), and O-shaped (‘O’). The X-axis represents 11 pure tone frequencies, and the Y-axis represents 6 sound intensities.



**Reviewer #2 (Public Review):**

*Summary:*

*Gu and Liang et. al investigated how auditory information is mapped and transformed as it enters and exits an auditory cortex. They use anterograde transsynaptic tracers to label and perform calcium imaging of thalamorecipient neurons in A1 and retrograde tracers to label and perform calcium imaging of corticothalamic output neurons. They demonstrate a degradation of tonotopic organization from the input to output neurons.*

*Strengths:*

*The experiments appear well executed, well described, and analyzed.*

*Weaknesses:*

*(1) Given that the CT and TR neurons were imaged at different depths, the question as to whether or not these differences could otherwise be explained by layer-specific differences is still not 100% resolved. Control measurements would be needed either by recording (1) CT neurons in upper layers, (2) TR in deeper layers, (3) non-CT in deeper layers and/or (4) non-TR in upper layers.*

We appreciate these constructive suggestions. To address this, we performed new experiments and analyses.

Comparison of TR neurons across superficial layers: we analyzed our existing TR neuron dataset to see if response properties varied by depth within the superficial layers. We found no significant differences in the fraction of tuned neurons, field IQR, or maximum bandwidth (BWmax) between TR neurons in L2/3 and L4. This suggests a degree of functional

homogeneity within the thalamorecipient population across these layers. The results are presented in new supplemental figures (Figure 2- figure supplementary 4).

Necessary control experiments.

(1) CT neurons in upper layers. CT neurons are thalamic projection neurons that only exist in the deeper cortex, so CT neurons do not exist in upper layers (Antunes and Malmierca, 2021).

(2) TR neurons in deeper layers. As we mentioned in the manuscript, due to high-titer AAV1-Cre virus labeling controversy (anterograde and retrograde labelling both exist), it is challenging to identify TR neurons in deeper layers.

(3) non-CT in deeper layers and/or (4) non-TR in upper layers.

To directly test if projection identity confers distinct functional properties within the same cortical layers, we performed the crucial control of comparing TR neurons to their neighboring non-TR neurons. We injected AAV1-Cre in MGB and a Cre-dependent mCherry into A1 to label TR neurons red. We then co-injected AAV-CaMKII-GCaMP6s to label the general excitatory population green. In merged images, this allowed us to functionally image and directly compare TR neurons (yellow) and adjacent non-TR neurons (green). We separately recorded the responses of these neurons to pure tones using two-photon imaging. The results show that TR neurons are significantly more likely to be tuned to pure tones than their neighboring non-TR excitatory neurons. This finding provides direct evidence that a neuron's long-range connectivity, and not just its laminar location, is a key determinant of its response properties. The results are presented in new supplemental figures (Figure 2- figure supplementary 5).

Related publications:

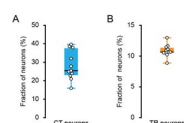
Antunes FM, Malmierca MS. 2021. Corticothalamic Pathways in Auditory Processing: Recent Advances and Insights From Other Sensory Systems. *Front Neural Circuits* 15: 721186. DOI: <https://doi.org/10.3389/fncir.2021.721186>, PMID: 34489648

(2) What percent of the neurons at the depths are CT neurons? Similar questions for TR neurons?

We thank the reviewer for the comments. We performed histological analysis on brain slices from our experimental animals to quantify the density of these projection-specific populations. Our analysis reveals that CT neurons constitute approximately 25.47%–36.50% of all neurons in Layer 6 of A1. In the superficial layers (L2/3 and L4), TR neurons comprise approximately 10.66%–11.37% of the total neuronal population.

## Author response image 2.

The fraction of CT and TR neurons. (A) Boxplots showing the fraction of CT neurons. N = 11 slices from 4 mice. (B) Boxplots showing the fraction of TR neurons. N = 11 slices from 4 mice.



(3) V-shaped, I-shaped, or O-shaped is not an intuitively understood nomenclature, consider changing. Further, the x/y axis for Figure 4a is not labeled, so it's not clear what the heat maps are supposed to represent.

The terms "V-shaped," "I-shaped," and "O-shaped" are an established nomenclature in the auditory neuroscience literature for describing frequency response areas (FRAs), and we use them for consistency with prior work. V-shaped: Neurons whose FRAs show decreasing frequency selectivity with increasing intensity. I-shaped: Neurons whose FRAs show constant frequency selectivity with increasing intensity. O-shaped: Neurons responsive to a small range of intensities and frequencies, with the peak response not occurring at the highest intensity level.

(Rothschild et al., 2010). We have included a more detailed description in the Methods.

The X-axis represents 11 pure tone frequencies, and the Y-axis represents 6 sound intensities. So, the heat map represents the FRA of neurons in A1, reflecting the responses for different frequencies and intensities of sound stimuli. In the revised manuscript, we have provided clarifications in the figure legend.

*(4) Many references about projection neurons and cortical circuits are based on studies from visual or somatosensory cortex. Auditory cortex organization is not necessarily the same as other sensory areas. Auditory cortex references should be used specifically, and not sources reporting on S1, and V1.*

We thank the reviewers for their valuable comments. We have made a concerted effort to ensure that claims about cortical circuit organization are supported by findings specifically from the auditory cortex wherever possible, strengthening the focus and specificity of our discussion.

#### **Reviewer #3 (Public Review):**

##### *Summary:*

*The authors performed wide-field and 2-photon imaging in vivo in awake head-fixed mice, to compare receptive fields and tonotopic organization in thalamocortical recipient (TR) neurons vs corticothalamic (CT) neurons of mouse auditory cortex. TR neurons were found in all cortical layers while CT neurons were restricted to layer 6. The TR neurons at nominal depths of 200-400 microns have a remarkable degree of tonotopy (as good if not better than tonotopic maps reported by multiunit recordings). In contrast, CT neurons were very heterogenous in terms of their best frequency (BF), even when focusing on the low vs high-frequency regions of the primary auditory cortex. CT neurons also had wider tuning.*

##### *Strengths:*

*This is a thorough examination using modern methods, helping to resolve a question in the field with projection-specific mapping.*

##### *Weaknesses:*

*There are some limitations due to the methods, and it's unclear what the importance of these responses are outside of behavioral context or measured at single timepoints given the plasticity, context-dependence, and receptive field 'drift' that can occur in the cortex.*

*(1) Probably the biggest conceptual difficulty I have with the paper is comparing these results to past studies mapping auditory cortex topography, mainly due to differences in methods. Conventionally, the tonotopic organization is observed for characteristic frequency maps (not best frequency maps), as tuning precision degrades and the best frequency can shift as sound intensity increases. The authors used six attenuation levels (30-80 dB SPL) and reported that the background noise of the 2-photon scope is <30 dB*

*SPL, which seems very quiet. The authors should at least describe the sound-proofing they used to get the noise level that low, and some sense of noise across the 2-40 kHz frequency range would be nice as a supplementary figure. It also remains unclear just what the 2-photon dF/F response represents in terms of spikes. Classic mapping using single-unit or multi-unit electrodes might be sensitive to single spikes (as might be emitted at characteristic frequency), but this might not be as obvious for Ca<sup>2+</sup> imaging. This isn't a concern for the internal comparison here between TR and CT cells as conditions are similar, but is a concern for relating the tonotopy or lack thereof reported here to other studies.*

We sincerely thank the reviewer for the thoughtful evaluation of our manuscript and for your positive assessment of our work.

#### (1) Concern regarding Best Frequency (BF) vs. Characteristic Frequency (CF)

Our use of BF, defined as the frequency eliciting the highest response averaged across all sound levels, is a standard and practical approach in 2-photon Ca<sup>2+</sup> imaging studies. (Issa et al., 2014; Rothschild et al., 2010; Schmitt et al., 2023; Tischbirek et al., 2019). This method is well-suited for functionally characterizing large numbers of neurons simultaneously, where determining a precise firing threshold for each individual cell can be challenging.

#### (2) Concern regarding background noise of the 2-photon setup

We have expanded the Methods section ("Auditory stimulation") to include a detailed description of the sound-attenuation strategies used during the experiments. The use of a custom-built, double-walled sound-proof enclosure lined with wedge-shaped acoustic foam was implemented to significantly reduce external noise interference. These strategies ensured that auditory stimuli were delivered under highly controlled, low-noise conditions, thereby enhancing the reliability and accuracy of the neural response measurements obtained throughout the study.

#### (3) Concern regarding the relationship between dF/F and spikes

While Ca<sup>2+</sup> signals are an indirect and filtered representation of spiking activity, they are a powerful tool for assessing the functional properties of genetically-defined cell populations. As you note, the properties and limitations of Ca<sup>2+</sup> imaging apply equally to both the TR and CT neuron groups we recorded. Therefore, the profound difference we observed—a clear tonotopic gradient in one population and a lack thereof in the other—is a robust biological finding and not a methodological artifact.

Related publications:

- (1) Issa JB, Haeffele BD, Agarwal A, Bergles DE, Young ED, Yue DT. 2014. Multiscale optical Ca<sup>2+</sup> imaging of tonal organization in mouse auditory cortex. *Neuron* 83: 944-59. DIO: <https://doi.org/10.1016/j.neuron.2014.07.009>, PMID: 25088366
- (2) Rothschild G, Nelken I, Mizrahi A. 2010. Functional organization and population dynamics in the mouse primary auditory cortex. *Nat Neurosci* 13: 353-60. DIO: <https://doi.org/10.1038/nn.2484>, PMID: 20118927
- (3) Schmitt TTX, Andrea KMA, Wadle SL, Hirtz JJ. 2023. Distinct topographic organization and network activity patterns of corticocollicular neurons within layer 5 auditory cortex. *Front Neural Circuits* 17: 1210057. DIO: <https://doi.org/10.3389/fncir.2023.1210057>, PMID: 37521334
- (4) Tischbirek CH, Noda T, Tohmi M, Birkner A, Nelken I, Konnerth A. 2019. In Vivo Functional Mapping of a Cortical Column at Single-Neuron Resolution. *Cell Rep* 27: 1319-1326 e5. DIO: <https://doi.org/10.1016/j.celrep.2019.04.007>, PMID: 31042460

*(2) It seems a bit peculiar that while 2721 CT neurons (N=10 mice) were imaged, less than half as many TR cells were imaged (n=1041 cells from N=5 mice). I would have expected there to be many more TR neurons even mouse for mouse (normalizing by number of neurons per mouse), but perhaps the authors were just interested in a comparison data set and not being as thorough or complete with the TR imaging?*

As shown in the Figure 2- figure supplementary 2, a much higher fraction of TR neurons was "tuned" to pure tones (46% of 1041 neurons) compared to CT neurons (only 18% of 2721 neurons). To obtain a statistically robust and comparable number of tuned neurons for our core analysis (481 tuned TR neurons vs. 491 tuned CT neurons), it was necessary to sample a larger total population of CT neurons, which required imaging from more animals.

*(3) The authors' definitions of neuronal response type in the methods need more quantitative detail. The authors state: "Irregular" neurons exhibited spontaneous activity with highly variable responses to sound stimulation. "Tuned" neurons were responsive neurons that demonstrated significant selectivity for certain stimuli. "Silent" neurons were defined as those that remained completely inactive during our recording period (> 30 min). For tuned neurons, the best frequency (BF) was defined as the sound frequency associated with the highest response averaged across all sound levels.". The authors need to define what their thresholds are for 'highly variable', 'significant', and 'completely inactive'. Is best frequency the most significant response, the global max (even if another stimulus evokes a very close amplitude response), etc.*

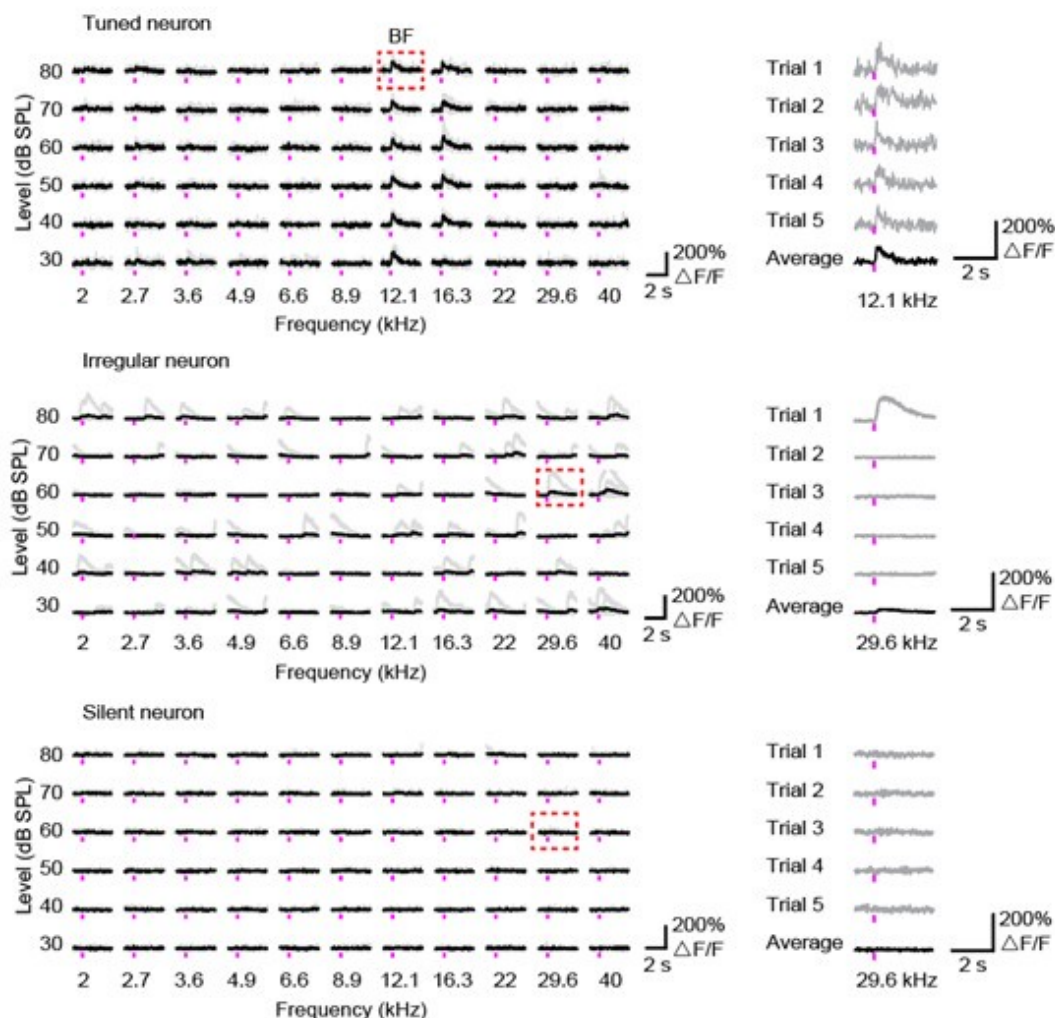
We appreciate the reviewer's suggestions. We have added more detailed description in the Methods.

**Tuned neurons:** A responsive neuron was further classified as "Tuned" if its responses showed significant frequency selectivity. We determined this using a one-way ANOVA on the neuron's response amplitudes across all tested frequencies (at the sound level that elicited the maximal response). If the ANOVA yielded a p-value < 0.05, the neuron was considered "Tuned". **Irregular neurons:** Responsive neurons that did not meet the statistical criterion for being "Tuned" (i.e., ANOVA p-value ≥ 0.05) were classified as "Irregular". This provides a clear, mutually exclusive category for sound-responsive but broadly-tuned or non-selective cells. **Silent neurons:** Neurons that were not responsive were classified as "Silent". This quantitatively defines them as cells that showed no significant stimulus-evoked activity during the entire recording session. **Best frequency (BF):** It is the frequency that elicited the maximal mean response, averaged across all sound levels.

To provide greater clarity, we showed examples in the following figures.



Author response image 3.



**Reviewer #1 (Recommendations For The Authors):**

(1) A1 and AuC were used exchangeably in the text.

Thank you for pointing out this issue. Our terminological strategy was to remain faithful to the original terms used in the literature we cite, where "AuC" is often used more broadly. In the revised manuscript, we have performed a careful edit to ensure that we use the specific term "A1" (primary auditory cortex) when describing our own results and recording locations, which were functionally and anatomically confirmed.

(2) Grammar mistakes throughout.

We are grateful for the reviewer's suggested improvement to our wording. The entire manuscript has undergone a thorough professional copyediting process to correct all grammatical errors and improve overall readability.

(3) The discussion should talk more about how/why L6 CT neurons don't possess the tonotopic organization and what are the implications. Currently, it only says 'indicative of an increase in synaptic integration during cortical processing'...

Thanks for this suggestion. We have substantially revised and expanded the Discussion section to explore the potential mechanisms and functional implications of the lack of tonotopy in L6 CT neurons.

**Broad pooling of inputs:** We propose that the lack of tonotopy is an active computation, not a passive degradation. CT neurons likely pool inputs from a wide range of upstream neurons with diverse frequency preferences. This broad synaptic integration, reflected in their wider tuning bandwidth, would actively erase the fine-grained frequency map in favor of creating a different kind of representation.

**A shift from topography to abstract representation:** This transformation away from a classic sensory map may be critical for the function of corticothalamic feedback. Instead of relaying "what" frequency was heard, the descending signal from CT neurons may convey more abstract, higher-order information, such as the behavioral relevance of a sound, predictions about upcoming sounds, or motor-related efference copy signals that are not inherently frequency-specific.

**Modulatory role of the descending pathway:** The descending A1-to-MGB pathway is often considered to be modulatory, shaping thalamic responses rather than driving them directly. A modulatory signal designed to globally adjust thalamic gain or selectivity may not require, and may even be hindered by, a fine-grained topographical organization.

**Reviewer #2 (Recommendations For The Authors):**

*(1) Given that the CT and TR neurons were imaged at different depths, the question as to whether or not these differences could otherwise be explained by layer-specific differences is still not 100% resolved. Control measurements would be needed either by recording (1) CT neurons in upper layers (2) TR in deeper layers (3) non-CT in deeper layers and/or (4) non-TR in upper layers.*

We appreciate these constructive suggestions. To address this, we performed new experiments and analyses.

**Comparison of TR neurons across superficial layers:** we analyzed our existing TR neuron dataset to see if response properties varied by depth within the superficial layers. We found no significant differences in the fraction of tuned neurons, field IQR, or maximum bandwidth (BWmax) between TR neurons in L2/3 and L4. This suggests a degree of functional homogeneity within the thalamorecipient population across these layers.

**Necessary control experiments.**

(1) CT neurons in upper layers. CT neurons are thalamic projection neurons that only exist in the deeper cortex, so CT neurons do not exist in upper layers (Antunes and Malmierca, 2021).

(2) TR neurons in deeper layers. As we mentioned in the manuscript, due to high-titer AAV1-Cre virus labeling controversy (anterograde and retrograde labelling both exist), it is challenging to identify TR neurons in deeper layers.

(3) non-CT in deeper layers and/or (4) non-TR in upper layers.

To directly test if projection identity confers distinct functional properties within the same cortical layers, we performed the crucial control of comparing TR neurons to their neighboring non-TR neurons. We injected AAV1-Cre in MGB and a Cre-dependent mCherry into A1 to label TR neurons red. We then co-injected AAV-CaMKII-GCaMP6s to label the general excitatory population green. In merged images, this allowed us to functionally image and directly compare TR neurons (yellow) and adjacent non-TR neurons (green). We

separately recorded the responses of these neurons to pure tones using two-photon imaging. The results show that TR neurons are significantly more likely to be tuned to pure tones than their neighboring non-TR excitatory neurons. This finding provides direct evidence that a neuron's long-range connectivity, and not just its laminar location, is a key determinant of its response properties.

Related publications:

Antunes FM, Malmierca MS. 2021. Corticothalamic Pathways in Auditory Processing: Recent Advances and Insights From Other Sensory Systems. *Front Neural Circuits* 15: 721186. DOI: <https://doi.org/10.3389/fncir.2021.721186>, PMID: 34489648

*(3) V-shaped, I-shaped, or O-shaped is not an intuitively understood nomenclature, consider changing. Further, the x/y axis for Figure 4a is not labeled, so it's not clear what the heat maps are supposed to represent.*

The terms "V-shaped," "I-shaped," and "O-shaped" are an established nomenclature in the auditory neuroscience literature for describing frequency response areas (FRAs), and we use them for consistency with prior work. V-shaped: Neurons whose FRAs show decreasing frequency selectivity with increasing intensity. I-shaped: Neurons whose FRAs show constant frequency selectivity with increasing intensity. O-shaped: Neurons responsive to a small range of intensities and frequencies, with the peak response not occurring at the highest intensity level.

(Rothschild et al., 2010). We have included a more detailed description in the Methods.

The X-axis represents 11 pure tone frequencies, and the Y-axis represents 6 sound intensities. So, the heat map represents the FRA of neurons in A1, reflecting the responses for different frequencies and intensities of sound stimuli. In the revised manuscript, we have provided clarifications in the figure legend.

*(4) Many references about projection neurons and cortical circuits are based on studies from visual or somatosensory cortex. Auditory cortex organization is not necessarily the same as other sensory areas. Auditory cortex references should be used specifically, and not sources reporting on S1, V1.*

We thank the reviewers for their valuable comments. We have made a concerted effort to ensure that claims about cortical circuit organization are supported by findings specifically from the auditory cortex wherever possible, strengthening the focus and specificity of our discussion.

**Reviewer #3 (Recommendations For The Authors):**

*I suggest showing some more examples of how different neurons and receptive field properties were quantified and statistically analyzed. Especially in Figure 4, but really throughout.*

We thank the reviewer for this valuable suggestion. To provide greater clarity, we have added more examples in the following figure.

<https://doi.org/10.7554/eLife.99989.2.sa0>